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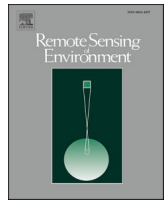

Quality control and crop characterization framework for multi-temporal UAV LiDAR data over mechanized agricultural fields

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ABSTRACT

Recent developments in remote sensing are enabling automatic, high resolution, and non-destructive survey of agriculture fields, providing the key basis for advancing plant breeding. Among the used remote sensing modalities, LiDAR has attracted wide attention for its ability to directly provide accurate 3D information. Despite the increasing utilization of LiDAR technology in phenotyping, there is still a lack of effective quality control strategies, in particular, quality control of LiDAR data collected on a multi-temporal basis. This study proposes a targetless framework for multi-temporal LiDAR data quality control and crop characterization in mechanized agricultural fields. Features extracted from the fields – terrain patches and row/alley locations – are utilized for evaluating the vertical and planimetric relative accuracy of the point clouds. Row/alley locations in the field are automatically identified from the point clouds based on the assumption that higher point density and/or higher elevation correspond to plant locations. The performance of the proposed quality control strategies is evaluated using multi-temporal datasets collected in agricultural fields of different sizes, orientation, crops, and growth stages. The result shows that the net vertical and planimetric discrepancies between multi-temporal point clouds are ±3 cm and ±8 cm, respectively. While the former reflects the actual accuracy of the point clouds, the latter is a combined effect of the LiDAR point cloud accuracy, rasterization artifacts, crop type, growth pattern, and wind condition during data acquisition. In terms of row and alley detection, the result shows that the proposed strategy achieves high performance and can deal with different planting orientation, crop types, growth stages, canopy cover, and planting density. In conclusion, this study presents a quality control framework for multi-temporal LiDAR data. Moreover, the row and alley detection leads to automated extraction of plots, and hence facilitates the use of remotely sensed data for automated phenotyping.

1. Introduction

Plant phenotyping provides plant traits required for analyzing the impact of genetic and environmental factors on the plant's behavior throughout the growth cycle. To take full advantage of modern biotechnology (e.g., DNA sequencing and editing), the ability to efficiently collect phenotypic data with high accuracy and high spatial and temporal resolution in a non-invasive manner is the key to advancing plant breeding. Effective, field-based, and high-throughput phenotyping was made possible by recent advances in remote sensing, including sensors, platforms, and methodologies for data processing and analysis (Weiss et al., 2020). Mobile platforms like unmanned aerial vehicles (UAVs) and ground-based systems can carry a variety of sensors and maneuver in the field to capture high-quality data. Imaging systems with different spectral ranges, including Red-Green-Blue (RGB) cameras, multispectral/hyperspectral cameras, thermal cameras, and chlorophyll fluorescence cameras are often used in plant phenotyping (Araus et al., 2018). These sensors can be used to detect and collect plant morphological data such as plant height, canopy cover, vegetation indices, leaf area index, leaf color, and growth stage (Ahmed et al., 2019; Chen et al., 2019; Maimaitijiang et al., 2020; Miao et al., 2020; Milioto et al., 2018; Ribera et al., 2018; Xu et al., 2019). High-throughput phenotyping using LiDAR, an active sensing technology that directly provides 3D coordinates, is an increasingly popular alternative for traditional field-based phenotyping. While achieving comparable performance for plant height estimation, LiDAR was reported to outperform image-based techniques in terms of the ability to capture ground points and estimation of physical biomass parameters (Sofonia et al., 2019). Several studies have developed LiDAR-based mobile platforms for field phenotyping (Malambo et al., 2019; Pérez-ruiz et al., 2020; Qiu et al., 2019;

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Sofonia et al., 2019; Sun et al., 2018; Zhou et al., 2020). A mobile LiDAR system requires direct georeferencing of the platform, i.e., trajectory information should be provided by an integrated survey-grade global navigation satellite systems/inertial navigation systems (GNSS/INS). The high cost for the procurement of these systems has been the limiting factor for using LiDAR for in-field phenotyping. Today, relatively low-cost LiDAR and direct georeferencing sensors are commercially available, which makes a mobile LiDAR system an affordable and increasingly popular option in field phenotyping.

To accurately quantify plant growth throughout the growing season, LiDAR data collected across multiple dates must be in agreement. However, despite the increased usage of LiDAR technology in phenotyping, there is still a lack of effective quality control (QC) strategies for multi-temporal LiDAR data in agricultural fields. Existing strategies for direct cloud-to-cloud comparison relies on well-distributed, stable surfaces with different orientation (i.e., different slop and aspect), which cannot be assumed in agricultural fields. For feature-based quality assessment strategies, a major challenge is the identification of conjugate features in overlapping point clouds (Habib et al., 2010). In urban areas, planar and linear features extracted from physical infrastructure (e.g., pavement surface, building roofs/facades, traffic signs, and utility poles) are appropriate and reliable QC features. In agricultural fields, however, it is quite difficult to identify planar or linear features because of the lack of man-made fixtures. Moreover, the observed surfaces change constantly as the plant grows, which increases the difficulty in identifying conjugate features. Using man-made targets is impractical because it is expensive, limited in terms of coverage, and usually poor in terms of spatial distribution especially for large agricultural fields. In most studies, an external accuracy assessment is used for quality control of the LiDAR data, for example, comparing LiDAR-based height information and manually measured plant heights (Sun et al., 2018; Zhou et al., 2020). Point cloud alignment between several dates, in contrast, has been neglected even though multi-temporal data acquisition which provides the required information about the plants throughout the growing season is the main feature of high-throughput phenotyping.

The focus of this study is quality control of multi-temporal LiDAR data as well as crop characterization. The proposed quality control framework evaluates the relative accuracy of multi-temporal LiDAR point clouds using features automatically identified and extracted from the field (terrain patches and row/alley locations, as illustrated in Fig. 1) without relying on manually deployed targets. The proposed approach targets mechanized agricultural fields, where a planter with auto-steer capability is used to sow seeds in straight rows and thus leads to a regular planting pattern. Our main contributions can be summarized as follows:

- We propose a quality control framework for the evaluation of relative accuracy of multi-temporal LiDAR data in mechanized agricultural fields. The approach does not involve target deployment.
- We develop a strategy for automated detection of rows and alleys in the field. This information serves as a precursor for plot extraction, and therefore facilitates the analysis of remotely sensed data for automated phenotyping.
- We evaluate the performance of the proposed quality control framework using eight datasets over four sites, with agricultural fields of different sizes, crops, planting patterns, planting densities, and growth stages.

The rest of the paper is structured as follows: Section 2 discusses recent work in field phenotyping and quality control of point cloud data. Section 3 describes the data acquisition system and field surveys. Section 4 focuses on the proposed quality control framework and row/alley detection approach. Section 5 presents the experimental results, followed by Section 6 that discusses the key findings. Finally, Section 7 provides conclusions and potential directions for future work.

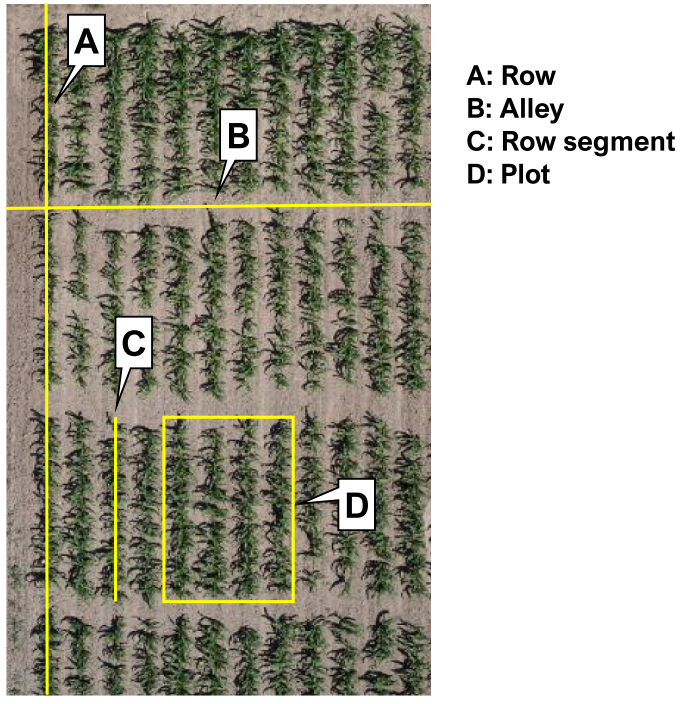

Fig. 1. The definition of row, alley, row segment, and plot (using a four-row plot as an example).

2. Related work

In this section, recent developments in field phenotyping using mobile mapping systems are first discussed. Then, we focus on literature dealing with the evaluation of the quality of LiDAR data with more emphasis on captured data over agricultural fields.

2.1. Automated phenotyping with mobile mapping systems

The development of strategies and algorithms to process, analyze, and extract information from the raw data collected in the fields is critical for high-throughput phenotyping. Researchers have developed various methods for extracting plant traits from imagery. For example, various studies focused on segmenting individual plots (which usually represent different treatments or genotypes), plants, or plant organs such as stalk, leaf, or panicle, and extracting plant traits including canopy cover, plant height, vegetation indices, leaf area index, plant and leaf morphology, plant and leaf color, and growth stage (Ahmed et al., 2019; Chen et al., 2019; Maimaitijiang et al., 2020; Miao et al., 2020; Milioto et al., 2018; Xu et al., 2019; Ribera et al., 2017; Ribera et al., 2018). To automatically extract individual plots from orthorectified images, image segmentation techniques based on Hue-Saturation-Value (HSV) color space (Ribera et al., 2017) and random walker algorithm (Ahmed et al., 2019) were adopted. More recently, researchers have used deep neural networks to customize appropriate features (i.e., informative and non-redundant values extracted from raw data) for different tasks such as biomass prediction (Masjedi et al., 2019). Nevertheless, a major limitation for imagery data is that it only provides planimetric information, and image-based 3D reconstruction techniques are sensitive to environmental factors (Guisado-Pintado et al., 2019; Lin et al., 2019; Mancini et al., 2013). The repetitive and low-texture patterns in agricultural field images further impede the performance of image-based 3D reconstruction techniques, and this issue is a current focus of the photogrammetric and computer vision fields (Hasheminasab et al., 2020).

The use of LiDAR is relatively new in agricultural applications. Su et al. (2019) used terrestrial LiDAR to collect plant morphological data

for maize, including height, plant area index, plant area density, and leaf area of each plant. Individual plants were manually segmented based on the prior information that the maize was planted in a rectangular cell at low density. Malambo et al. (2019) derived panicle counts and sizes in a sorghum field using height, point density, and color information acquired by a terrestrial LiDAR. The LiDAR was equipped with an RGB camera to produce color-coded point clouds. The method worked on a plot-level, which has been established manually using pre-defined plot boundaries. The phenotypic data for individual sorghum plants were collected by manual measurements from the point clouds. The analysis was limited to plots closest to the scanners since the point density decreases significantly as the plant-to-scanner distance increases. Sun et al. (2018) used 2D LiDAR onboard a wheel-based mobile platform to measure height, canopy cover, and plant volume of cotton plots. Multi-date LiDAR point clouds were used to monitor plant traits throughout the growing season to detect growth patterns of different cotton cultivars. Pérez-ruiz et al. (2020) developed a wheel-based mobile system to measure canopy height in a wheat field. The system carried a LiDAR and multiple spectral reflectance sensors, and used an odometry system for geolocation. The canopy height was estimated by analyzing the heights and number of points in longitudinal sections extracted from a manually-defined region of interest. Qiu et al. (2019) developed a wheel-based mobile system equipped with a 3D LiDAR for phenotyping maize. Their robot collected data in a stop-and-go manner: the robot moved to a pre-determined location, stopped and scanned, and then moved to the next stop. The scans from different spots were then merged through post-acquisition registration. Row spacings were detected via analyzing the point cloud distribution histogram, with the assumption that the stalk of a plant should produce denser LiDAR points than the leaves. Plant heights were then computed based on the detected crop row locations. Zhou et al. (2020) used multi-temporal LiDAR point clouds acquired by a UAV-based system to monitor changes in plant heights of lodged maize. The canopy height was obtained using a commercial software by first, generating a digital terrain model (DTM) and a digital surface model (DSM) and then, subtracting the DTM from the DSM.

The above discussion reveals that LiDAR provides accurate 3D information that allows for the extraction of multiple features describing the geometric characteristics of plants. Height and point density (distribution of the points) are relevant information for deriving plant characteristics and are used by most researchers. Prior research also verified that LiDAR-based measurements including plant height, projected canopy area, plant volume/area, and panicle dimension are highly correlated with ground truth data (Sun et al., 2018; Malambo et al., 2019; Su et al., 2019). Because of the irregular nature of the LiDAR point clouds, analyzing LiDAR data is much more complicated than analyzing image data. Currently, the analysis using LiDAR data requires impractical data acquisition procedure and a large amount of tedious manual intervention.

2.2. Quality control of point cloud data

Quality control refers to a post-mission procedure for ensuring data integrity, correctness, and completeness (Habib et al., 2010). Quality control typically involves evaluating two types of accuracy: i) absolute/external accuracy: the agreement between the point cloud and independently measured ground control points/targets, and ii) relative/internal accuracy: the alignment between overlapping point clouds collected at different dates. Strategies for evaluating the agreement between two point clouds can be divided into two categories: point cloud-based and feature-based. The most commonly-used point cloud-based approach is the iterative closest point (ICP) (Besl and McKay, 1992). The ICP assumes that closest points from two point clouds constitute a conjugate point pair, and uses the point pair for discrepancy estimation. This approach is simple, fast, and does not require gridding or meshing of the datasets. However, it is sensitive to the point density and noise in the point cloud. Several attempts were made to improve the robustness of the ICP using local neighborhood characteristics (Sharp et al., 2002; Gressin et al., 2013). Variants of ICP, such as the Multiscale Model to Model Cloud Comparison (M3C2), estimate the discrepancy using automatically-identified solid surfaces in the point clouds (Lague et al., 2013). The M3C2 approach assumes that the area of interest has both horizontal and vertical solid surfaces, and evaluates the discrepancy between two point clouds along the surface normal. The original manuscript recommends vegetation removal prior to discrepancy estimation since the different degrees of penetration within the point clouds over vegetated areas would not allow for representing those areas as 3D manifold surfaces. For the ICP and its variants, reliable discrepancy estimation depends on well-distributed stable surfaces with different orientation (i.e., different slop and aspect). For instance, the M3C2 approach has been mainly introduced for geomorphological applications such as landslide and rock-fall monitoring, where one can assume that stable surfaces with varying orientation are available. For agriculture applications, on the other hand, the only stable area is the terrain. Thus, relatively horizontal terrain cannot provide sufficient information for estimating the planimetric discrepancy, which is the most visible artifact due to biases in the system calibration parameters (Habib et al., 2010). Therefore, the ICP, M3C2, and their derivatives cannot be used for checking the co-alignment of multi-temporal point clouds covering agricultural fields. The other group of algorithms evaluates the agreement of two point clouds using conjugate features extracted from the point clouds. Habib et al. (2010) proposed the use of conjugate planar/linear features, and conjugate point-patch pairs for evaluating LiDAR data quality. These features can be extracted reliably from point clouds in most urban and rural areas. In agricultural fields, however, planar and linear features with different orientation are not directly available. Therefore, appropriate features must be established/identified in order to perform reliable quality control over agricultural fields.

Most of the studies that utilized mobile LiDAR for collecting field phenotypic data conducted external accuracy assessment of LiDAR-derived plant heights using manual field measurements (Sun et al., 2018; Zhou et al., 2020). LiDAR-derived plant height was shown to be accurate, which verifies the quality of the point cloud. Sofonia et al. (2019) compared the performance of UAV LiDAR and photogrammetry for monitoring sugarcane growth. The LiDAR and image-based point clouds achieved comparable performance in estimating plant heights, capturing the canopy throughout the growing season. LiDAR data outperformed image-based point cloud in mapping below canopy features throughout the complete growth cycle. Nevertheless, to the best of the authors' knowledge, there is no study that investigates the quality of LiDAR data collected throughout the growing season. The absence of reliable quality control limits our ability to fully utilize multi-temporal datasets, limiting the potential of using LiDAR data to extract more information related to plant characteristics.

3. Data acquisition system and case studies

LiDAR and imagery data were collected using a custom-built, UAV-based mobile mapping system over four sites. This section introduces the system architecture and calibration strategy. Further, we describe the study sites and provide information regarding the flight configuration for field surveys.

3.1. System description and calibration

The UAV-based mobile mapping system used in this study is shown in Fig. 2. The system consists of a DJI M600 Pro hexacopter, which carries a Velodyne VLP-16 Puck Lite laser scanner, Sony α7R III RGB camera, Headwall Nano-Hyperspec VNIR push-broom hyperspectral scanner, and Trimble APX-15 UAV v3 integrated GNSS/INS unit. This study focuses on the quality control of LiDAR data. The RGB camera data is only utilized to generate an orthophoto, which is used to visualize the

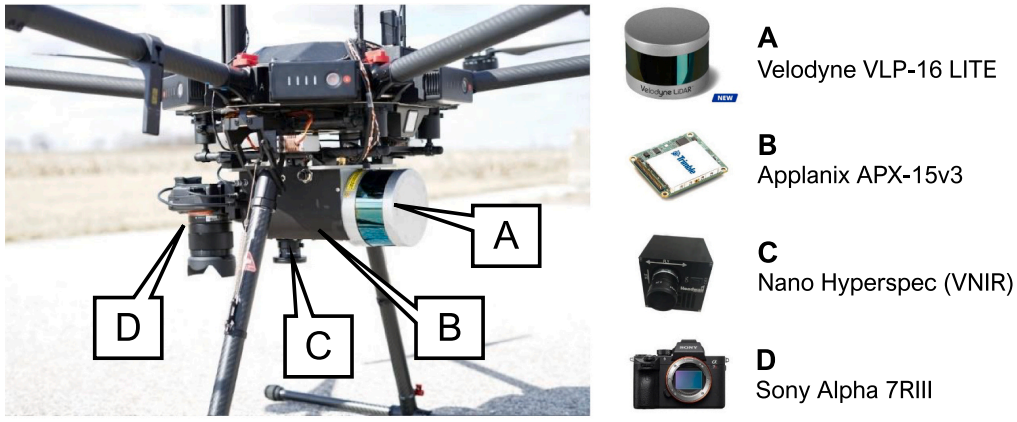

Fig. 2. The Unmanned Aerial Vehicle (UAV)-based mobile mapping system and onboard sensors used in this study.

detected row/alley locations, and the hyperspectral data is not used. The GNSS/INS unit provides direct georeferencing information (i.e., the position and attitude information of the vehicle frame at 200 Hz). The expected post-processing positional accuracy is 2 to 5 cm, and the attitude accuracy is 0.025° and 0.08° for the roll/pitch and heading, respectively. The VLP-16 scanner has 16 radially oriented laser rangefinders that are aligned vertically from +15° to −15°, leading to a total vertical field of view (FOV) of 30°, and it rotates internally to achieve a 360° horizontal FOV. The scanner captures around 300,000 points per second, with a range accuracy of ±3 cm and a maximum range of 100 m (Velodyne, 2019). The Sony α7R III has an image resolution of 42 MP and is triggered by an Arduino Micro microcontroller board to capture images at 1.5 s interval.

For mobile mapping systems utilizing direct georeferencing, a system calibration that determines the relative position and orientation between the onboard sensors and GNSS/INS unit (denoted hereafter as mounting parameters) is crucial to produce accurate georeferenced products. In this study, the mounting parameters were rigorously determined through the calibration approach introduced in Ravi et al. (2018a, 2018b). The FOV across the flying direction was set to ±70° from nadir. The expected accuracy of the reconstructed point cloud was derived according to the accuracy of individual sensors and standard deviations of the mounting parameters (estimated from system calibration) using the LiDAR Error Propagation calculator introduced by Habib et al. (2006). The horizontal and vertical accuracy at nadir was within a 5 to 6 cm range at a flying height of 50 m. At the edge of the swath, the horizontal accuracy was within an 8 to 9 cm range, and the vertical accuracy was still within a 5 to 6 cm range.

3.2. Datasets description

A total of eight datasets, as listed in Table 1, collected over four sites are used for assessing the accuracy of multi-temporal LiDAR data and the performance of the proposed row/alley detection strategy. Figs. 3 and 4 show the location and sample orthophoto together with the corresponding trajectory of the UAV (shown in black) for each site/field. Information regarding field trials and flight configurations can be found in Table 1.

Datasets A-1 and A-2 are used for evaluating the absolute accuracy of the LiDAR point clouds. The datasets were collected over a test site – an open area near the Indiana Corn and Soybean Innovation Center (ICSC) building at the Purdue Agronomy Center for Research and Education (ACRE) in Indiana, USA. Twelve highly reflective checkboard targets were deployed on the site. The yellow boxes in Fig. 4a show the locations of the twelve checkboard targets. The ground coordinates of the center of these checkerboard targets were surveyed using a Real-Time Kinematic-Global Navigation Satellite Systems (RTK-GNSS) survey with a Trimble R10 GNSS receiver. The survey was conducted using an InCORS (a network of continuously operating reference stations managed by the Indiana Department of Transportation) base-station. Also, the UTM coordinate system with NAD83 as the datum is adopted in this study. Considering the distance between the rover receiver and the GNSS base-station, i.e., 6 km, the expected horizontal and vertical accuracy from the RTK-GNSS survey is in the range of ±2 to ±3 cm and ±3 to ±4 cm, respectively (Trimble, 2018).

Datasets B-1, B-2, B-3, C-1, and C-2 are used to evaluate the relative accuracy of multi-temporal LiDAR point clouds, as well as the performance of the proposed row/alley detection strategy. The field surveys were carried out in two fields of ACRE (denoted as ACRE-21C and ACRE-42, hereafter) during the summer of 2019. The two fields were both oriented in the North-South direction, as shown in the orthophotos (Fig. 4b and c). A planter was used for sowing in rows. The row separation of the planter was 30 in. (0.762 m). ACRE-21C was used for popcorn (maize) breeding trials to test how different popcorn hybrids respond to standard and high planting density. The field consisted of 90 rows with a row segment length of 3.5 m. ACRE-42 was used for sorghum breeding trials. The panel had 630 hybrids replicated twice in four row plots. The field consisted of 260 rows with a row segment length of

Table 1
Field information and flight configuration for the datasets. Datasets A-1 and A-2 were collected over a test site for absolute accuracy evaluation.

ID	Site/field	Crop	Sowing date	Growth stage	Data collection date	Flying height (m)	Ground speed (m/s)	Lateral distance (m)
A-1	ICSC	N/A	N/A	N/A	3/4/2020	41	4.0	6
A-2					5/1/2020	41	4.0	6
B-1	ACRE-21C	Popcorn	6/3/2019	V8-V10[a]	7/12/2019	30	3.4	9
B-2		(maize)		VT-R1[a]	8/10/2019	45	4.2	9
B-3				R3[a]	9/4/2019	44	4.1	9
C-1	ACRE-42	Sorghum	6/4/2019	Floral initiation[b]	7/12/2019	46	5.0	13
C-2				Flowering [b]	8/10/2019	47	5.0	13
D-1	Patancheru	Maize	12/4/2019	R0[a]	2/18/2020	38	3.8	8

[a] Based on the standard described in Abendroth et al. (2011).
[b] Based on the standard provided by the Agricultural Production Systems sIMulator (APSIM) model (APSIM, 2020).

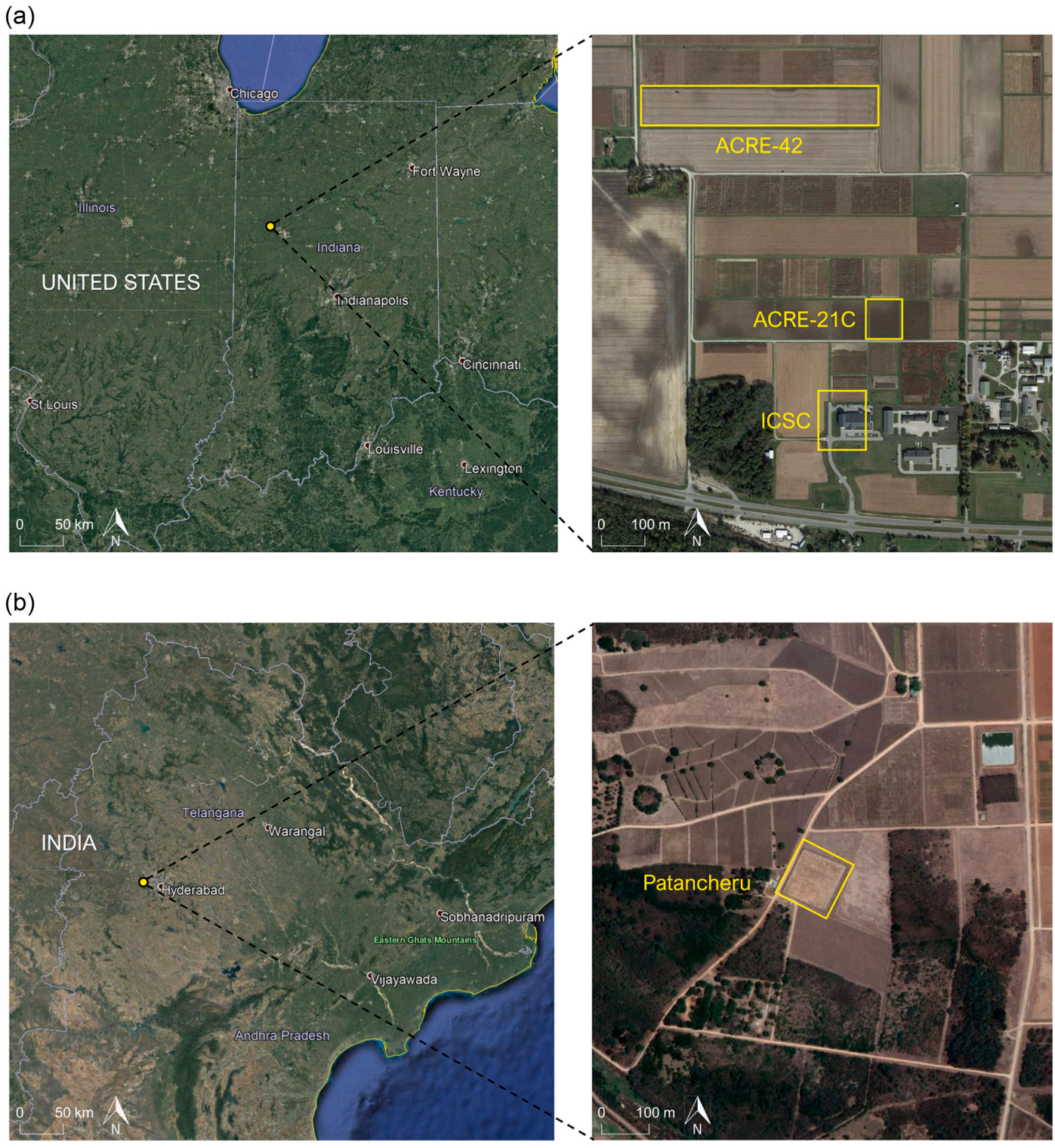

Fig. 3. Locations of the study sites (a) in Indiana, USA, and (b) in Patancheru, India (aerial photo adapted from Google Earth Images).

3.5 m.

Dataset D-1 was acquired over a field at Patancheru, India during the spring of 2020. This dataset is used to test the capability of row/alley detection in a manually-planted field with a different planting orientation and planting density. As can be seen in the orthophoto (Fig. 4d), the field is approximately oriented in a Northeast-Southwest direction. Moreover, the row and alley directions are not orthogonal. The field was used for maize breeding trials for heat stress tolerance, and was divided into two parts. The first part consisted of 62 rows in one row plot with a row segment length of 2.5 m. The second part consisted of 73 rows in one/two row plots with a row segment length of 3 m. The crops were planted manually using planting rulers.

4. Methodology for LIDAR quality control and row/alley detection

Quality control involves evaluating i) relative accuracy: alignment between multi-temporal point clouds, and ii) absolute accuracy: agreement between the point cloud and independently measured ground control points/targets. The assessment of relative and absolute accuracy

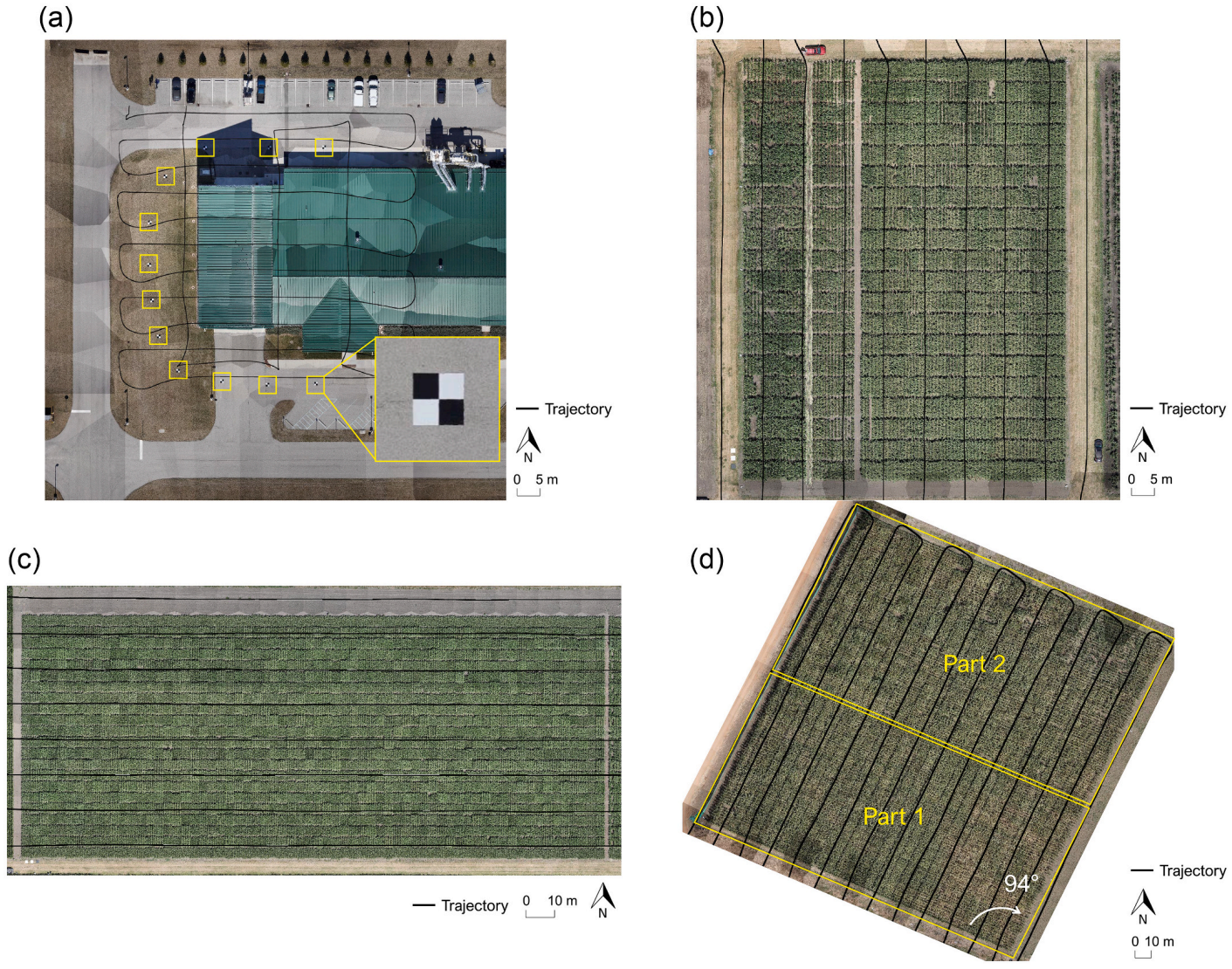

Fig. 4. Sample orthophotos and trajectory (black lines) showing: (a) ICSC with checkerboard targets for dataset A-1, (b) ACRE-21C for dataset B-1, (c) ACRE-42 for dataset C-2, and (d) Patancheru for dataset D-1.

of multi-temporal point clouds is based on evaluating the degree of consistency among conjugate points/features. Conceptually, conjugate features in overlapping point clouds should match each other well if there are no systematic and/or random errors. In other words, any systematic discrepancies between two point clouds would signify the presence of biases in the LiDAR measurements, GNSS/INS trajectory, and/or mounting parameters. The proposed quality control strategy evaluates the x, y, and z shifts between overlapping point clouds. To achieve targetless quality control, two features are automatically identified in agricultural plantations: terrain patches are used to evaluate the relative vertical accuracy, and rows/alleys for relative planimetric accuracy. An independent RTK-GNSS survey is conducted for the assessment of the absolute accuracy of the point clouds.

A flowchart of the proposed quality control framework is shown in Fig. 5. This section explains each component in detail. Starting with point cloud reconstruction, Section 4.1 presents the ground filtering and DSM generation techniques adopted in this study. Next, the identification of conjugate points/features used to evaluate different types of accuracy is introduced. Section 4.2 derives the corresponding LiDAR points for the RTK-GNSS measurements for absolute accuracy assessment. Section 4.3 introduces the planar features, terrain patches, for evaluating the relative vertical accuracy. Section 4.4 describes the detection of rows/alleys, the linear features used for evaluating the relative planimetric accuracy. Finally, the least squares model for estimating the net discrepancy between two point clouds using conjugate points/features is presented in Section 4.5.

4.1. Point cloud reconstruction, ground filtering, and DSM generation

Direct georeferencing is adopted in this study for processing LiDAR data. Combing the LiDAR range measurement and orientation of the laser beam establishes the coordinates of a point, I, captured at time t relative to the laser unit frame, $r_I^{lu(t)}$. The GNSS/INS integration provides the trajectory information, i.e., position, $r^m_{b(t)}$, and orientation, $R^m_{b(t)}$, of the GNSS/INS body frame relative to the mapping frame. The system calibration determines the lever arm, r^b_{lu}, and boresight matrix, R^b_{lu}, relating the laser unit frame to the GNSS/INS body frame. The 3D coordinates of the point, I, in the mapping frame can then be derived as per Eq. (1), which is graphically illustrated in Fig. 6 (El-Sheimy et al., 2005; Ravi et al., 2018a). Using the trajectory provided by the GNSS/INS unit, the LiDAR point clouds from different flight lines/dates are reconstructed in a common mapping frame. In this study, the UTM coordinate system with NAD83 as the datum is adopted. Therefore, there is no need for registration to align different flight lines. The quality of the point cloud depends on the accuracy of the LiDAR unit, trajectory, and system

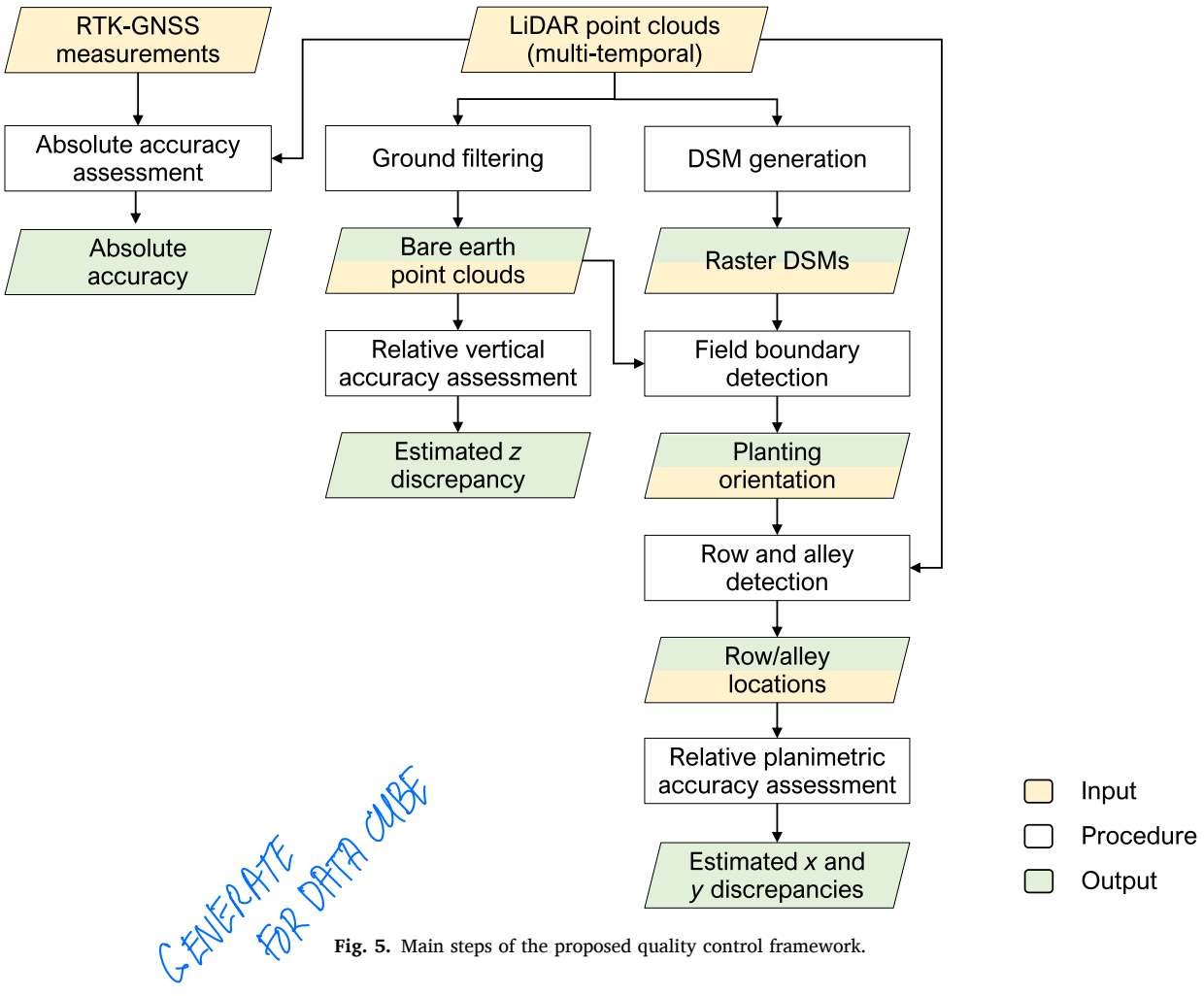

Fig. 5. Main steps of the proposed quality control framework.

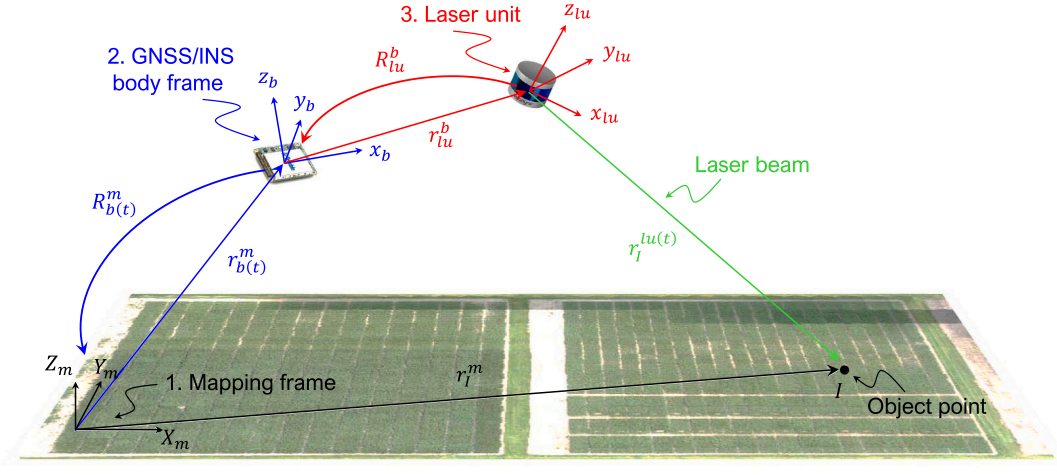

Fig. 6. Illustration of point positioning for a LiDAR system.

calibration (Ravi et al., 2018a, 2018b). For a reliably-calibrated LiDAR system and precise trajectory, the flight line configuration will not impact the point cloud quality. A previous study validated the quality of the point clouds derived from a similar system (Lin et al., 2019), where it was shown that the alignment between strips is within the noise level of the point cloud even at the edge of the swath.

$$r_I^m = r_{b(t)}^m + R_{b(t)}^m r_{lu}^b + R_{b(t)}^m R_{lu}^b r_I^{lu(t)} \quad (1)$$

Upon reconstructing the point cloud, a ground filtering algorithm is applied to separate bare earth points, which represent the terrain, and above-ground points. Cloth simulation filtering (Zhang et al., 2016) is adopted for its robustness and superior performance, which have been validated by several studies (Wan et al., 2018; Serifoglu Yilmaz et al., 2018). The conceptual basis of the cloth simulation approach can be summarized in four steps:

- Turn the point cloud upside down,

- Define a cloth (consisting of particles and their interconnections) with some rigidness, and place it above the inverted point cloud,
- Let the cloth drop under the influence of gravity and designate the final shape of the cloth as the DTM, and
- Use the DTM to filter ground from non-ground points.

In this study, the height threshold for filtering ground points is set to 0.1 m, considering that the expected vertical accuracy of the point clouds is in the ±8−9 cm range. One should note that the bare earth point cloud is a subset of the point cloud, not a rasterized terrain model, and thus retains the precision of the point cloud.

Finally, a uniform gridded surface model (i.e., DSM) is generated based on the approach described in Lin et al., (2019). The approach uses the 90th percentile height to represent the surface to mitigate the noise impact. It also applies a nearest neighbor interpolation and a median filter to fill empty DSM cells and smooth the surface. For retaining the inherent level of spatial information, the DSM resolution should be selected based on the overall point density. In this study, the reconstructed point cloud had an average point density higher than 1400 points/m^2, which corresponds to an inter-point spacing of approximately 3 cm. The DSM resolution was set to 5 cm to ensure that the majority of the cells would retain elevation information from actual data rather than interpolated values.

4.2. Absolute accuracy assessment using RTK-GNSS measurements

The absolute accuracy of the point clouds is assessed against independently collected RTK-GNSS measurements of checkerboard targets. Conducting a reliable comparison between RTK-GNSS measurements and LiDAR point clouds is contingent on establishing point correspondences. To derive the coordinates of the center of checkerboard targets from a LiDAR point cloud, the following strategy is adopted. First, an initial approximation of the center of a checkerboard target is manually identified based on the intensity information. Since the checkerboard targets have high reflective coating, the intensity values on the targets will be much higher than the surrounding background. Next, all the points that lie within a spherical neighborhood of the initial manually-selected point are extracted. The radius of the spherical neighborhood is pre-defined to be 0.5 m since the actual width of the checkerboard target is 0.75 m. This radius ensures that the points within the neighborhood are mostly on the checkerboard target, and the neighborhood encompass the true center of the checkerboard target. The extracted points would thus form a planar segment. Parameters of the plane (Habib and Lin, 2016) are estimated through multiple iterations of plane fitting followed by outlier removal based on the root-mean-square error (RMSE) of the fit (hereafter denoted as iterative plane fitting). Iterations are performed until no more outliers can be detected. The LiDAR-derived center of the checkerboard target is defined as the projection of the initial point on the best-fitted plane for the neighborhood. The coordinate differences between the RTK-GNSS measurements and corresponding LiDAR-derived centers of the checkerboard targets, $\begin{bmatrix} d_{x_obs} & d_{y_obs} & d_{z_obs} \end{bmatrix}^T$, are finally evaluated.

4.3. Relative vertical accuracy assessment using terrain patches

In spite of the absence of man-made objects and elevation change of the surface throughout the growing season, change in the terrain is marginal. Furthermore, as the agricultural fields are typically flat or have mild slope, the terrain can be approximated as a plane in a local neighborhood. Consequently, as long as some of the LiDAR points can penetrate through canopy to capture points along the ground, and the ground filtering algorithm can effectively extract such bare earth point cloud, the terrain patches will be appropriate planar features for relative vertical quality control in agricultural fields.

Starting with two bare earth point clouds acquired at different dates, one is selected as a reference and another as a source. The reference and source point clouds are then segmented into patches with a given size (see Fig. 7). A principal component analysis (PCA) is carried out to determine whether a patch is a planar, linear, or rough feature, as proposed in Habib and Lin (2016). Only planar patches are included in subsequent evaluations. Iterative plane fitting is performed to estimate the parameters of a plane. A patch is rejected from the accuracy assessment if: i) the plane-fitting RMSE exceeds a user-defined threshold, or ii) the remaining points after iterative plane fitting fails to reach another user-defined threshold. Next, the center of the reference plane (a_i) and its projection on the source plane (b_i) are determined. The discrepancy between the conjugate patches would therefore be quantified by the coordinate difference between a_i and b_i (as depicted in Fig. 7). This discrepancy, $\begin{bmatrix} d_{x_obs} & d_{y_obs} & d_{z_obs} \end{bmatrix}^T$, can be expressed by the normal distance of a_i from the source plane (nd), and normal vector of the source plane (w_i), as shown in Eq. (2).

$$\begin{bmatrix} d_{x_obs} \\ d_{y_obs} \\ d_{z_obs} \end{bmatrix} = nd \begin{bmatrix} w_x \\ w_y \\ w_z \end{bmatrix} \tag{2}$$

4.4. Relative planimetric accuracy assessment using row/alley locations

The features used for relative planimetric quality control are the rows and alleys as the plants are the most distinct feature in the field, and their planimetric locations remain the same throughout the growing season. Assuming that crops are planted along straight lines and the length of row segments are the same, the rows/alleys in the xy plane can be represented as straight lines. The detection of rows and alleys is based on the hypothesis that a location with higher point density and/or higher elevation would correspond to plant location (see Fig. 8). The point clouds captured by UAV LiDAR are distinctly different from those captured by static LiDAR. For static LiDAR, scanner to object distance is the main factor that affects the point density. For UAV LiDAR, the object space starts from the canopy surface and goes down to the terrain. Since the canopy height is much smaller relative to the flying height, the point density along the scan lines depends more on the vegetation density. Although the majority of the LiDAR points will capture the top of the canopy in agricultural plantations, LiDAR can penetrate through the canopy surface and capture plant organs such as stalks and leaves below the canopy. Therefore, it is plausible to detect plant locations using the point density.

The proposed row/alley detection approach begins with planting

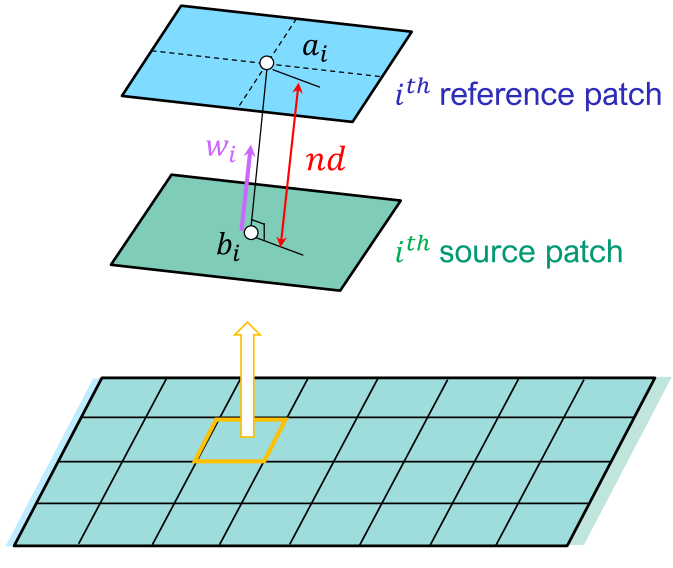

Fig. 7. Illustration of the extraction of terrain planar patches and the discrepancy estimation using conjugate terrain patches.

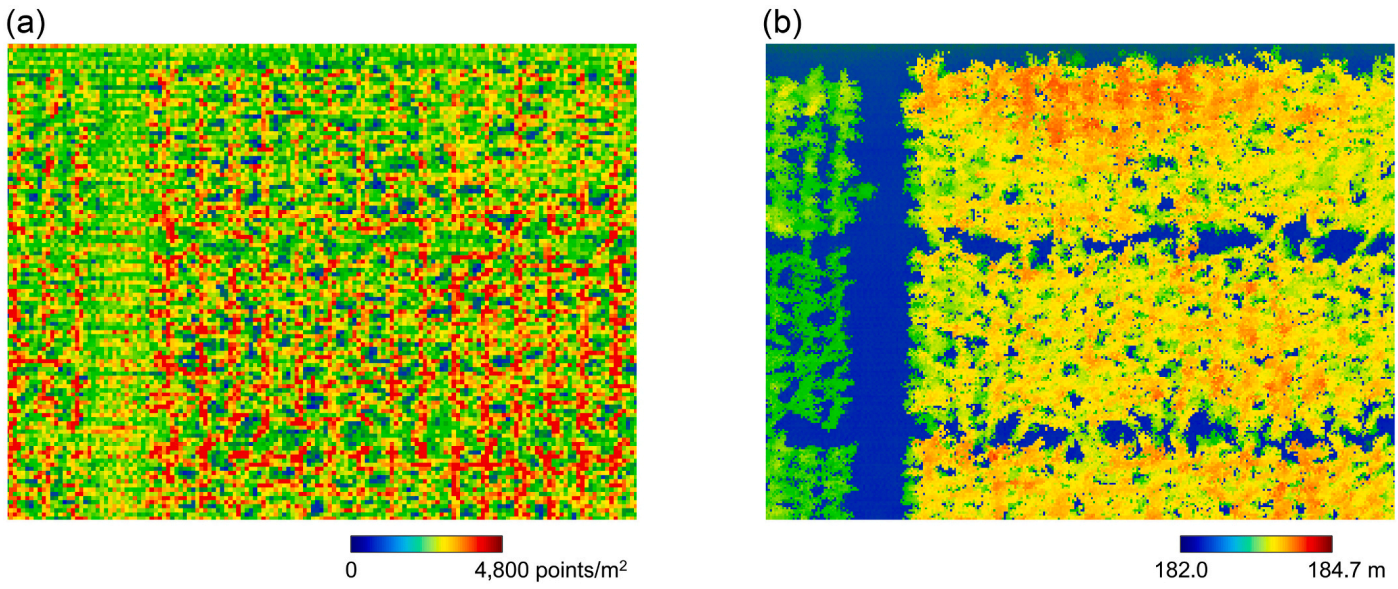

Fig. 8. Sample (a) point density and (b) elevation maps showing a portion of the field using the B-2 dataset.

orientation identification. The DSM represents the top of the canopy, the bare earth point cloud depicts the ground, and the elevation difference between the two indicates the plant height within the field. Sudden changes in this estimated plant height would define edges of the field (ground-to-crop transitions) or edges between plots (height differences between crop varieties). These edges are expected to be linear features along the rows and/or alleys. The segmentation approach proposed by Habib and Lin (2016) is applied to segment the 2D linear features and estimate the line parameters. The azimuth for rows' and alleys' direction are calculated based on the estimated line parameters. Fig. 9 shows an example of the DSM, detected field boundary, and line segments using the B-3 dataset.

Upon determining the planting orientation, the point cloud is rotated to a local coordinate system (uv) so that the row/alley is along the v/u axis (assuming the rows and alleys are approximately orthogonal). Two-dimensional cells are created along the uv plane over the field. The following three metrics are defined for each cell: i) the number of points in the cell, ii) the sum of elevations of all points in the cell, and iii) the 90^{th} percentile elevation of the cell. The first and third quantities reflect the point density and height of the point cloud within a local area, respectively, while the second quantity is a combination of the two characteristics. Next, the column sum and row sum of each metric (M) are computed as per Eqs. (3) and (4), where m and n denote the number of rows and columns within the created 2D cells, respectively.

$$Column\ sum = \sum_{v=1}^{n} M(u,v) \tag{3}$$

$$Row\ sum = \sum_{u=1}^{m} M(u,v) \tag{4}$$

The local peaks of the column sum indicate the local maxima of the point density and/or height, which would correspond to the row locations. Similarly, the local valleys of the row sum would specify the locations of the alleys. The peak/valley detection is based on finding the local maxima/minima by comparing of neighboring values. Two parameters are used for peak/valley detection: minimum horizontal distance between two peaks/valleys, and minimum prominence of a peak/valley. The prominence is a measure of how much a peak/valley stands out from the surrounding baseline (as illustrated in Fig. 10). In this study, minimum horizontal distance between two peaks/valleys is given according to the prior information of row spacing and row segment length. The minimum prominence of a peak/valley needs to be tuned for each dataset since it is related to the point density, which depends highly on technical factors pertaining to data acquisition. Fig. 11 shows an example of row/alley detection in two dates. As mentioned in the introduction of Section 4, the proposed quality control framework

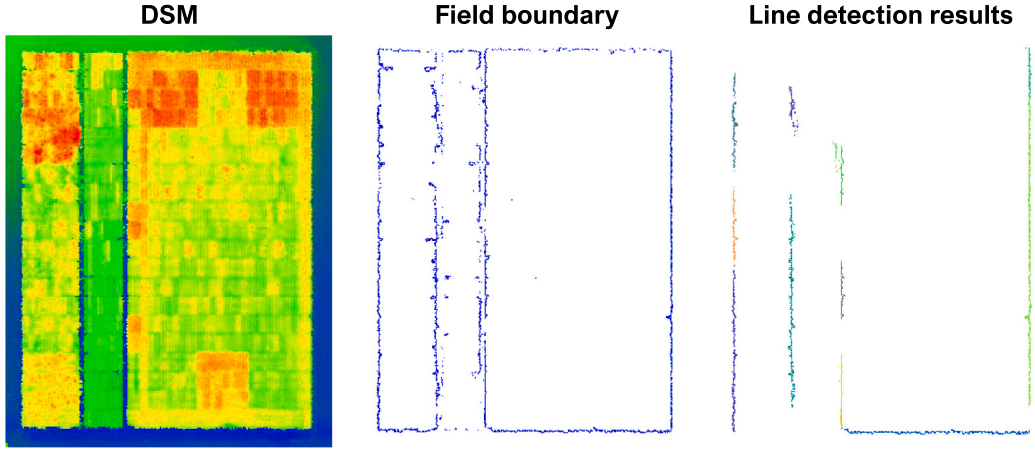

Fig. 9. Sample of DSM, detected field boundary, and detected line segments using the B-3 dataset.

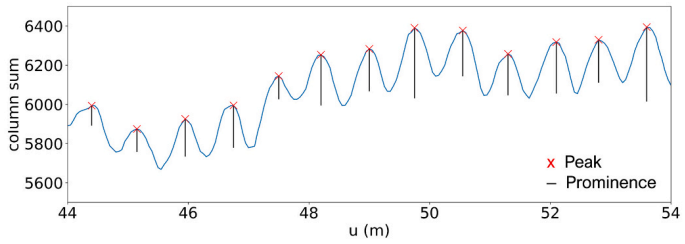

Fig. 10. Illustration of the prominences (the black lines) of the peaks using the D-1 dataset.

evaluates only the net shift between two point clouds. Assuming the orientation of a row/alley does not change over time, its location can be represented by the intersection of the row/alley and the u/v axis. That is, the location of a row, a, and an alley, b, in the local coordinate system can be written as $a(u_a, 0)$ and $b(0, v_b)$, respectively (as shown in Fig. 11). One should note that the above-mentioned strategy can deal with non-orthogonal rows and alleys by applying different rotation to the point cloud so that the row/alley is parallel to the v/u axis, and detecting rows/alleys independently.

Once the row and alley locations are detected in each dataset, the actual row spacing and row segment length in the field can be estimated by calculating the average distance between successive rows/alleys. In this study, the estimated row spacing is compared against the row separation of the planter as an external quality control. Conjugate rows and alleys among multi-temporal point clouds are paired by searching for the nearest row/alley within a given search radius. The search radius is defined by the estimated row spacing/row segment length. Finally, the row/alley locations are transformed to the mapping frame coordinate system, $a(x_a, y_a)$ and $b(x_b, y_b)$, using the rotation matrix from the uv local coordinate system to the mapping frame coordinate system, R_{uv}^{xy}. The planimetric discrepancy between conjugate rows/alleys in the mapping frame coordinate system, $\begin{bmatrix} d_{x_obs} & d_{y_obs} \end{bmatrix}^T$, can then be calculated.

To evaluate the capability of the three metrics – number of points, sum of elevation, and 90th percentile elevation – in terms of detecting rows/alleys, the row/alley detection results are compared against manually labeled ground truth. The locations of rows/alleys in the fields are manually identified using the orthophoto. A detection is classified as true positive (TP) if the distance between the detection and ground truth is smaller than a range threshold; otherwise, it is classified as false positive (FP). In this research, the range threshold is set to 0.25 m, which is one-third of the expected row spacing (0.762 m based on the row separation of the planter). A false negative (FN) denotes a case where a ground truth is present but the algorithm fails to detect it. To quantitatively evaluate the performance of each metric, precision, recall, and F1-score, as presented in Eqs. (5) to (7), are reported. Precision indicates how relevant the positive detections are while recall reveals how well the actual rows/alleys are identified. F1-score, a harmonic mean of precision and recall, is a measure of overall performance.

$$Precision = \frac{TP}{TP + FP} \tag{5}$$

$$Recall = \frac{TP}{TP + FN} \tag{6}$$

$$F_1 - score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{7}$$

4.5. Mathematical model for net discrepancy estimation

In this study, conjugate points (RTK-GNSS measurements and LiDAR-derived centers of the checkerboard targets), conjugate planar features (terrain patches), and conjugate linear features (rows/alleys) are used to evaluate the absolute accuracy, relative vertical accuracy, and relative planimetric accuracy of multi-temporal point clouds. The discrepancies between conjugate points, conjugate terrain patches, and conjugate rows/alleys ($\begin{bmatrix} d_{x_obs} & d_{y_obs} & d_{z_obs} \end{bmatrix}^T$) established in Sections 4.2, 4.3, and 4.4, respectively, are direct observations of the net discrepancy between two point clouds in the mapping frame ($\begin{bmatrix} d_x & d_y & d_z \end{bmatrix}^T$). Consequently, a least squares adjustment (LSA) model for estimating the net discrepancy between two point clouds can be written as per Eq. (8), where the random noise $\begin{bmatrix} e_x & e_y & e_z \end{bmatrix}^T$ follows a stochastic distribution with a zero mean and variance-covariance matrix Σ.

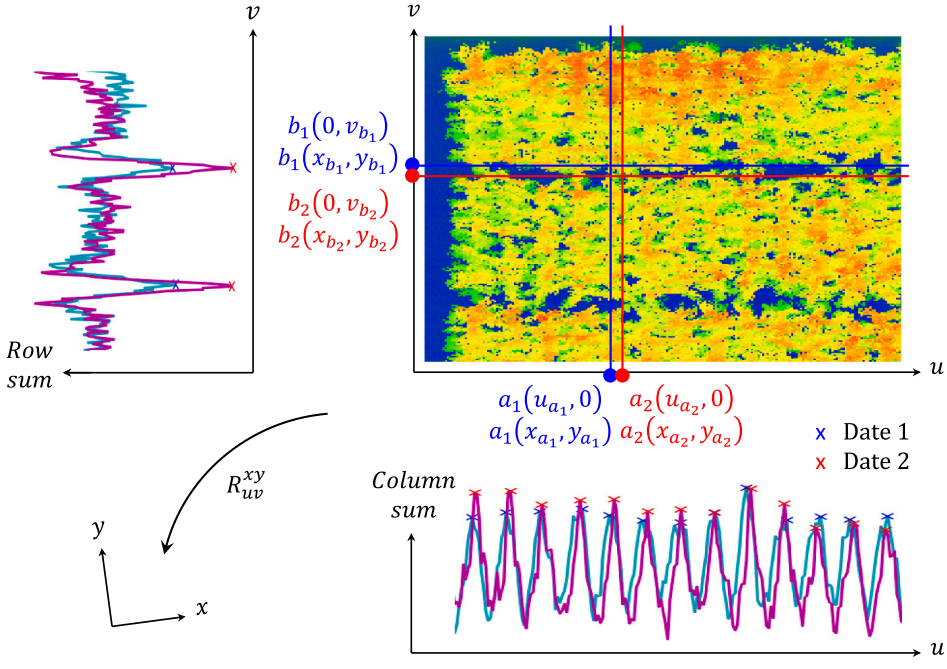

Fig. 11. Illustration of the coordinate system involved in row/alley detection, peak/alley detection results, conjugate rows (a_1 and a_2), and conjugate alleys (b_1 and b_2).

$$\begin{bmatrix} d_{x_obs} \\ d_{y_obs} \\ d_{z_obs} \end{bmatrix} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} + \begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix}, e^{\sim}(0, \Sigma) \tag{8}$$

For point-based pairing, each point pair provides information to estimate the discrepancy along *x*, *y*, and *z* directions. In this case, the solution of the LSA model is equal to reporting the mean, standard deviation, and RMSE of the observations. For feature-based pairing, each planar feature provides discrepancy information only along the normal direction to the plane, while each linear feature provides discrepancy information along any directions that are normal to the line direction. This leads to the need of incorporating a modified weight matrix in the LSA model. The concept of using weight modification to deal with feature-to-feature correspondence can be traced back to Renaudin et al. (2011). First, a local coordinate system (*uvw*) is defined. For planar features, the *u* and *v* axes are arbitrary, and the *w* axis is parallel to the plane normal (as illustrated by Fig. 12a). For linear features, the *u* axis is aligned along the line direction, and the *v* and *w* axes are arbitrary (as illustrated by Fig. 12b). The modified weight matrix for planar features in the local coordinate system is given by Eq. (9), where the weights in the two directions along the plane surface are set to zeros. For linear features, the modified weight matrix nullifies the discrepancy along the line direction, as shown in Eq. (10). This matrix is then transformed to the mapping frame coordinate system (*xyz*) based on Eq. (11), where R_{uvw}^{xyz} denotes the rotation matrix from the feature local coordinate system to the mapping frame coordinate system. It can be observed that the rank of the modified weight matrix for planar features and linear features is one and two, respectively. The discussion above shows a general 3D case. However, the proposed relative planimetric quality assessment approach only considers the 2D locations of the rows and alleys. In this case, the local coordinate system (*uv*) for linear features is defined as follows: the *u* axis is aligned along the line direction, and the *v* axis is perpendicular to the *u* axis. The modified weight matrix in the local coordinate system is given by Eq. (12), where the weight along the line direction is set to zero, leading to a rank of one.

$$P'_{uvw} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & P_w \end{bmatrix} \tag{9}$$

$$P'_{uvw} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & P_v & P_{vw} \\ 0 & P_{wv} & P_w \end{bmatrix} \tag{10}$$

$$P'_{xyz} = R_{uvw}^{xyz} P'_{uvw} R_{xyz}^{uvw} \tag{11}$$

$$P'_{uv} = \begin{bmatrix} 0 & 0 \\ 0 & P_v \end{bmatrix} \tag{12}$$

The least squares adjustment model with modified weight matrix for estimating the net discrepancy between two point clouds can be written as per Eq. (13). The random noise $\begin{bmatrix} e_x & e_y & e_z \end{bmatrix}^T$ has a mean of zero and variance-covariance matrix of $\sigma_0^2 P'^{+}_{xyz}$, with σ_0^2 is a-priori variance factor and P'_{xyz} is the modified weight matrix in the mapping frame. The plus sign denotes the Moore–Penrose pseudoinverse since P'_{xyz} is rank-deficient and its inverse does not exist. The least squares adjustment leads to the solution vector in Eq. (14), where *A* is the design matrix, which is an identity matrix in this case, and *y* is the vector of observations. The residuals, a-posteriori variance factor, and variance-covariance matrix can be computed as per Eqs. (15), (16) and (17), respectively.

$$\begin{bmatrix} d_{x_obs} \\ d_{y_obs} \\ d_{z_obs} \end{bmatrix} = \begin{bmatrix} d_x \\ d_y \\ d_z \end{bmatrix} + \begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix}, e^{\sim}\left(0, \sigma_0^2 P'^{+}_{xyz}\right) \tag{13}$$

$$\hat{x} = \begin{bmatrix} \hat{d}_x \\ \hat{d}_y \\ \hat{d}_z \end{bmatrix} = \left(A^T P' A\right)^{-1} A^T P' y = \left(\sum P'_i\right)^{-1} \sum P'_i \begin{bmatrix} d_{x_obs} \\ d_{y_obs} \\ d_{z_obs} \end{bmatrix}_i \tag{14}$$

$$\tilde{e}_i = \begin{bmatrix} e_x \\ e_y \\ e_z \end{bmatrix} = \begin{bmatrix} d_{x_obs} \\ d_{y_obs} \\ d_{z_obs} \end{bmatrix}_i - \begin{bmatrix} \hat{d}_x \\ \hat{d}_y \\ \hat{d}_z \end{bmatrix} \tag{15}$$

$$\hat{\sigma}_0^2 = \frac{\sum \tilde{e}_i^T P'_i \tilde{e}_i}{rank(P') - 3} \tag{16}$$

$$\hat{D}\begin{Bmatrix} \hat{d}_x \\ \hat{d}_y \\ \hat{d}_z \end{Bmatrix} = \hat{\sigma}_0^2 \left(\sum P'_i\right)^{-1} \tag{17}$$

5. Experimental results

In this section, the performance of the proposed quality control framework and the row/alley detection approach is evaluated. Theoretically, the estimated relative accuracy should be at the same level as the absolute accuracy in the absence of system-related biases (i.e., system calibration issues). Any inconsistencies between the two would indicate factors other than the point cloud accuracy (e.g., the cell size for row/alley detection, growth pattern of the crop, and wind condition) affecting the results. The absolute accuracy of the LiDAR point clouds from the UAV-based mobile mapping system was first assessed using the A-1 and A-2 datasets. Using the B-1, B-2, B-3, C-1, and C-2 datasets, the performance of the row/alley detection approach was evaluated. Then, the relative vertical and planimetric discrepancies between multi-temporal point clouds were estimated using the proposed quality control strategy, and compared against the absolute accuracy. Finally, the D-1 dataset was used to test the robustness of the row/alley detection strategy under non-orthogonal row/alley directions and different planting density.

5.1. Absolute quality control

The absolute accuracy of the point clouds was assessed against RTK-GNSS measurements of twelve checkerboard targets deployed on the site. The coordinates of the center of checkerboard targets were derived from LiDAR data by the approach described in Section 4.2. The coordinate differences between RTK-GNSS measurements and the corresponding LiDAR-derived centers of the checkerboard targets were

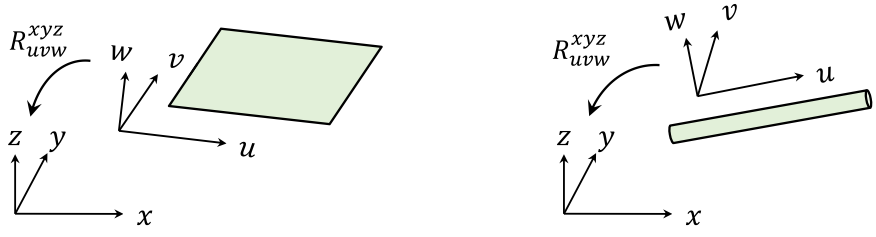

Fig. 12. Illustration of the relationship between the feature local coordinate system (*uvw*) and the mapping frame coordinate system (*xyz*) for: (a) a planar feature and (b) a linear feature.

calculated and reported in Table 2. The RMSE values in x, y, and z directions ranges from 2 to 4 cm for both datasets. The results reveal that the absolute accuracy of the point cloud from the UAV-based system is in the ±4 cm range, which is better than the expected accuracy of ±8–9 cm. The consistent results obtained for the different dates further verify the stability of the UAV-based system in terms of the resultant point cloud accuracy.

5.2. Bare earth point clouds and DSMs generation

Starting with the point cloud reconstruction, the bare earth point clouds were extracted using the cloth simulation, and the raster DSMs were generated for each dataset. Figs. 13 and 14 show the multi-temporal point clouds/DSMs in the maize fields (datasets B-1, B-2, and B-3) and the sorghum field (datasets C-1 and C-2), respectively. Although not specifically analyzed in this study, the point clouds we collected captured relevant information for agricultural applications. For example, the destructive sampling at location *i* can be observed in Fig. 13a, b, and c. A comparison of Fig. 14a and Fig. 14b shows that the sorghum had a similar height in July, but showed a large height variation among plots in August. This is because the plots were planted with hybrids that have different growth capacities. The raster DSMs display similar spatial patterns to the original point clouds but are smoothed and less noisy due to the rasterization and median filter. In contrast to the DSMs, which capture the change of the surface, the bare earth point clouds reveal that the terrain did not change throughout the growing season as the spatial patterns observed from different dates are similar. With the increase of canopy cover in the mid- and late-season, fewer LiDAR points are captured over the ground, resulting in gaps in the bare earth point clouds, and a decrease in the point density. For instance, the average point density for the C-1 and C-2 datasets is 1443 points/m^2 and 1489 points/m^2, respectively. For the bare earth point clouds, the average point density is 678 points/m^2 for the C-1 dataset, and 426 points/m^2 for the C-2 dataset. This suggests that fewer LiDAR points were able to penetrate to the ground as the canopy cover increases, to the point that more than two-thirds of the points are above-ground points.

To investigate the performance of the cloth simulation filtering in areas with dense canopy cover, a region of interest (ROI) consisting of three plots (see Fig. 14b and f) was manually cropped from the original and bare earth point clouds of the C-2 dataset. The cloth simulation filtering was able to correctly separate the bare earth point from the above-ground points, as can be observed in Fig. 14g. This observation is consistent across all the datasets, showing that even with low penetration, the cloth simulation filtering was able to identify the bare earth points, and the resultant terrain is reliable.

5.3. Row/alley detection

The performance of the row/alley detection approach was evaluated using the B-1, B-2, B-3, C-1, and C-2 datasets. First, the planting orientation of each field was determined. For both fields, the planting orientation was aligned in the North-South direction with the estimated azimuth being a negligible 0.45° and 0.40°, respectively. Therefore, there was no need to rotate the point clouds before proceeding to row/alley detection. Row/alley detection was carried out for each dataset using the strategy proposed in Section 4.4. The cell size for calculating the different metrics was set as 5 cm. The three metrics – number of points, sum of elevations, and 90th percentile elevation – were evaluated. Figs. 15 and 16 show the row/alley detection results using each metric in the maize fields (datasets B-1, B-2, and B-3) and the sorghum field (datasets C-1 and C-2), respectively. In general, all three metrics were able to produce clear peaks/valleys for row/alley detection. The destructive sampling in the B-1, B-2, and B-3 datasets can be observed (location *i* in Fig. 15b). The valleys based on the number of points metric in the B-1 dataset were less clear, which resulted in some error in the detection, as shown in location *ii* in Fig. 15c. This is related to the fact that the plants were small and the canopy was sparse in early-season, and therefore the number of points on the plants and the ground are similar. The result suggests that the LiDAR-based row/alley detection approach yields better performance in mid- and late-season, in contrast to the image-based plot segmentation approaches (Ahmed et al., 2019; Ribera et al., 2017) which performs better in early-season.

The performance of the three metrics – number of points, sum of elevations, and 90th percentile elevation – were evaluated using the B-1, B-2, B-3, C-1, and C-2 datasets, and reported in Table 3. Overall, all the metrics achieved high accuracy, supporting the observation that almost all the peaks/valleys can be identified using either one of the three metrics. The slightly lower recall score of 90th percentile elevation was mainly a result of the misdetection in the C-2 dataset, as depicted in location *i* in Fig. 16b. These less-prominent peaks were caused by the dense canopy cover and large height variation among plots. The sum of elevation provided the highest F1-score, and therefore deemed as the most robust metric in the experiments. This result is expected because the metric considers the point density and height, and both characteristics are related to plant location.

To demonstrate the row/alley detection result, we superimposed the detected row/alley locations as straight lines on top of the DSMs and orthophotos. Fig. 17 shows the results for portions of the fields using the B-2 and C-2 datasets, where the rows and alleys are detected using the sum of elevation metric. The detected rows and alleys are aligned well with the actual row centers and alleys shown in the DSMs and orthophotos. This observation is consistent through all the datasets, verifying the ability of the proposed algorithm to identify row/alley locations with different crops at different growth stages.

5.4. Relative vertical quality control

The net z discrepancies between multi-temporal point clouds collected over the maize fields (datasets B-1, B-2, and B-3) and the sorghum field (datasets C-1 and C-2) were evaluated using the planar features, terrain patches. Datasets B-1 and C-1 were selected as the reference. The terrain patches were first extracted from the bare earth point clouds. The discrepancies between conjugate patches were computed and used as the observations in the LSA model. The LSA estimated the net discrepancy between two point clouds, and reported the square root of a-posteriori variance factor ($\widehat{\sigma_0}$), an estimation of the overall net discrepancy while considering the modified weight matrix (as presented in Table 4). The square root of a-posteriori variance factor suggests the overall net discrepancy between the multi-temporal datasets is in the range of 3 cm, well below the established absolute accuracy of ±4 cm. The planimetric discrepancy (d_x and d_y) ranges from −12 cm to 11 cm, and the vertical discrepancy is in the range of ±3 cm. To evaluate the reliability of the discrepancies estimated by the least squares adjustment, a PCA analysis was conducted on the resultant dispersion matrix. The PCA analysis shows that the third principal component, which is almost parallel to the z direction, explains less than 0.1% of the total variance. On the contrary, the first and second principal components account for over 99.9% of the total variance, and the corresponding eigenvectors lie on the xy plane. This result implies that the z discrepancy estimation is the most reliable component among the

Table 2
Coordinate differences between the LiDAR-derived and RTK-GNSS coordinates of the center of the twelve checkerboard targets.

Dataset	Statistics	x differences (m)	y differences (m)	z differences (m)
A-1	Mean	0.03	0.01	0.03
	Std. Dev.	0.02	0.03	0.01
	RMSE	0.04	0.03	0.04
A-2	Mean	−0.01	0.00	0.01
	Std. Dev.	0.03	0.02	0.02
	RMSE	0.03	0.02	0.02

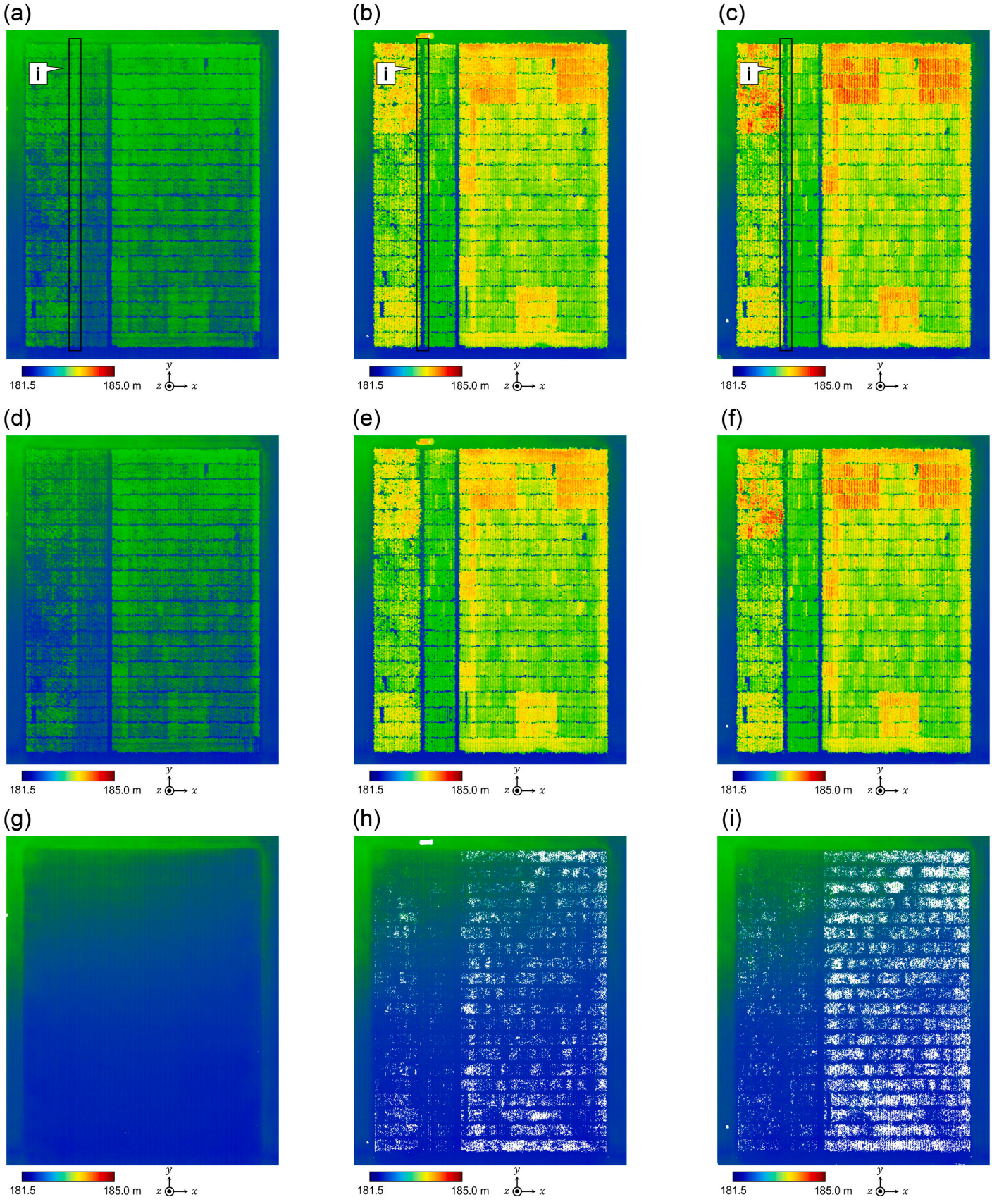

Fig. 13. LiDAR-based products including: point clouds for (a) B-1, (b) B-2, and (c) B-3 datasets where destructive sampling can be observed at location *i*; raster DSMs for (d) B-1, (e) B-2, and (f) B-3 datasets; bare earth point clouds for (g) B-1, (h) B-2, and (i) B-3 datasets. The point clouds/DSMs are colored by height and plotted with a white background.

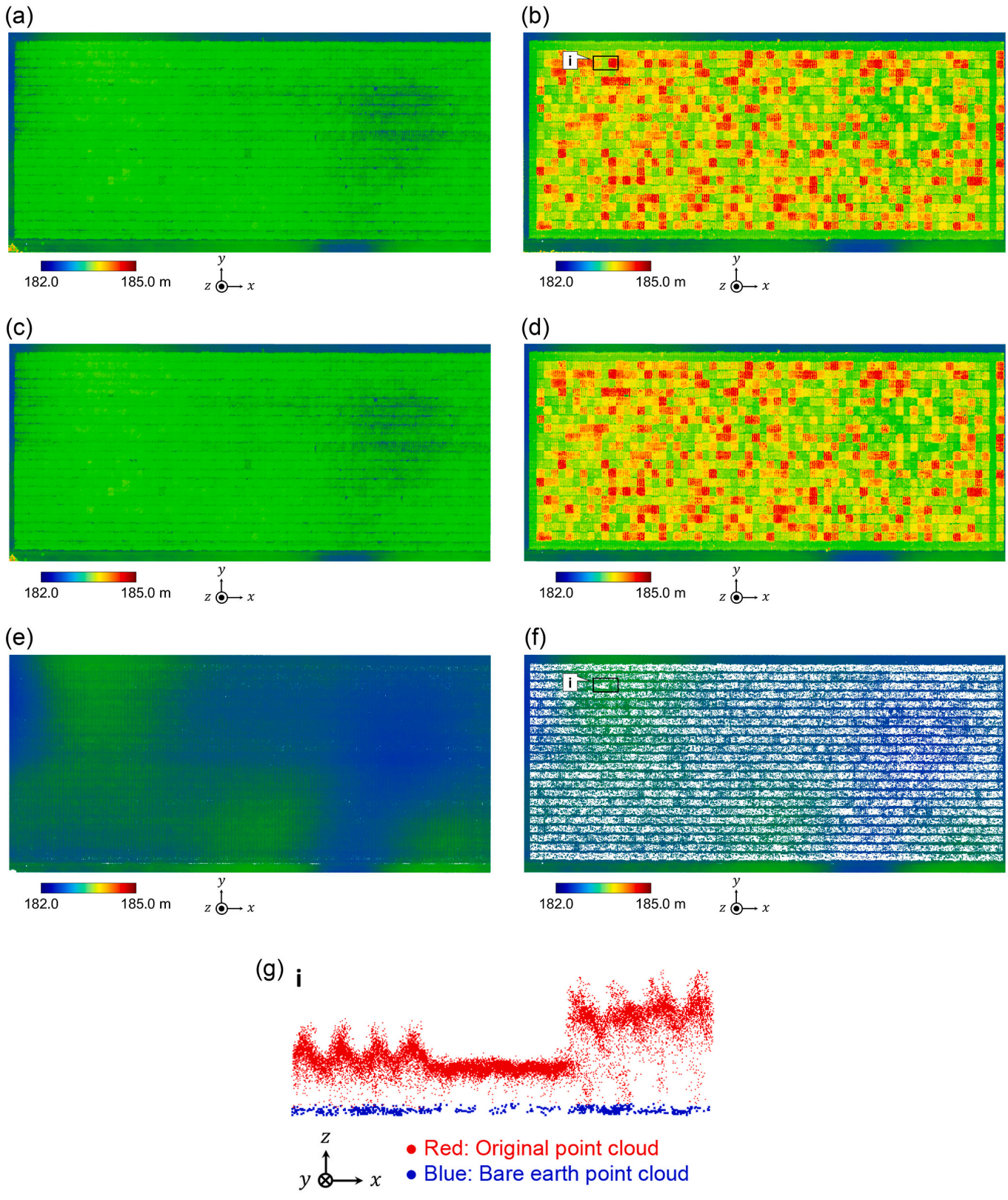

Fig. 14. LiDAR-based products including: point clouds for (a) C-1 and (b) C-2 datasets; raster DSMs for (c) C-1 and (d) C-2 datasets; bare earth point clouds for (e) C-1 and (f) C-2 datasets. The point clouds/DSMs are colored by height and plotted with a white background. A side view of the region at location *i*, showing (d) point cloud and bare earth point cloud extracted from the C-2 dataset.

three discrepancies. In contrast, the estimated *x* and *y* discrepancies do not reflect the actual accuracy of the point clouds due to the lack of variation of plane orientation. To conclude, the LSA suggests an overall net discrepancy of 3 cm, and the estimated vertical discrepancy is also in the range of ±3 cm. These discrepancies are within the range of the absolute accuracy of ±4 cm, verifying that the proposed relative vertical quality control approach can reflect the accuracy of the point clouds.

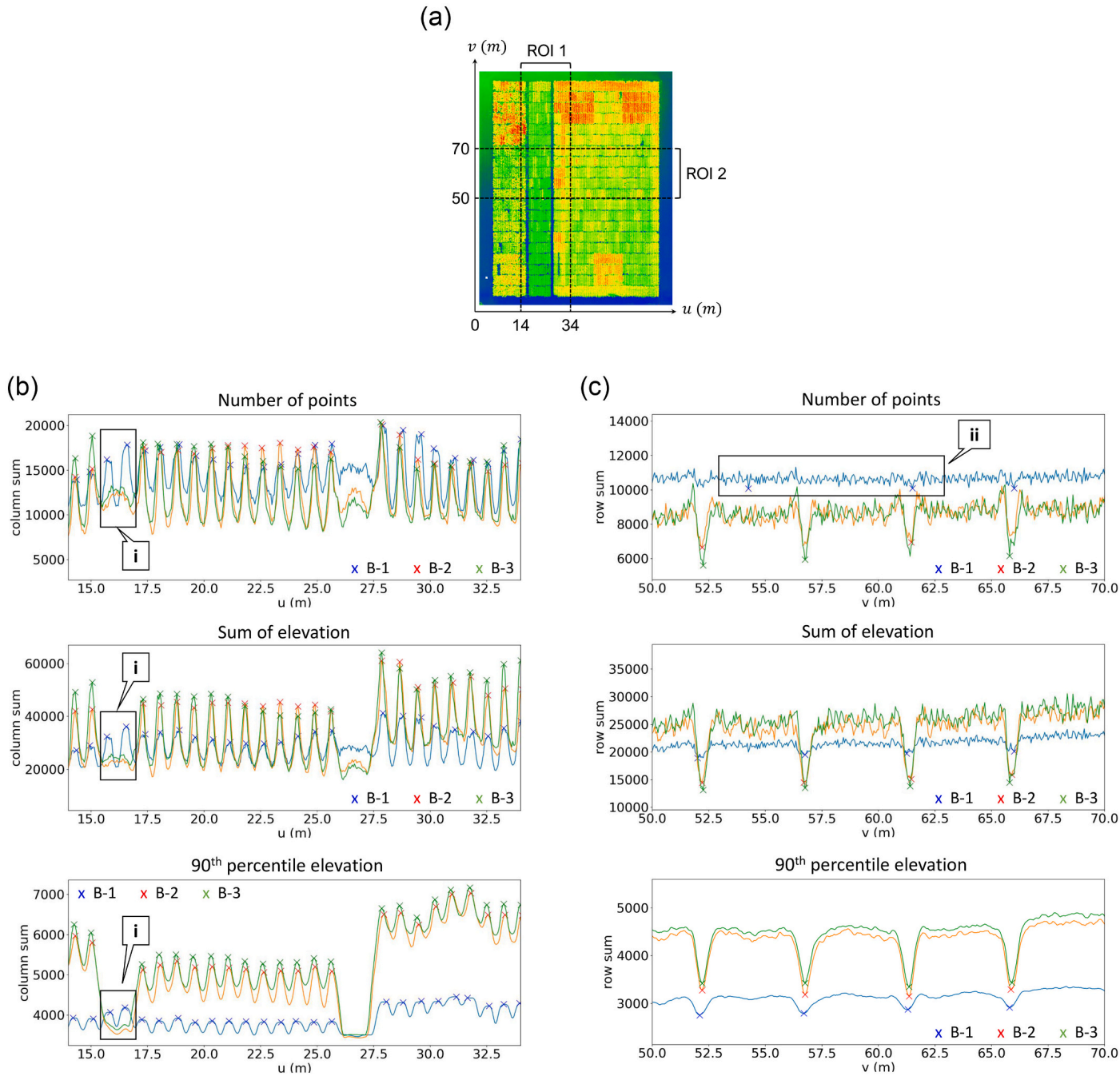

Fig. 15. Peak/valley detection with the three metrics for the B-1, B-2, and B-3 datasets: (a) point cloud and the selected ROIs (showing the B-3 dataset), (b) peak detection in ROI 1 where destructive sampling can be observed at location *i*, and (c) valley detection in ROI 2 where some error can be observed at location *ii*.

5.5. *Relative planimetric quality control*

The net *x* and *y* discrepancies between multi-temporal point clouds collected over the maize fields (datasets B-1, B-2, and B-3) and the sorghum field (datasets C-1 and C-2) were evaluated using the linear features defined by rows and alleys. Table 5 lists the net discrepancies and the square root of a-posteriori variance factor ($\widehat{\sigma_0}$) estimated using the three metrics. The square root of a-posteriori variance factor suggests that the overall net discrepancy between the multi-temporal datasets is in the range of 5 to 8 cm, which is higher than the absolute accuracy of ±4 cm. This result implies that factors other than the LiDAR accuracy affects the net discrepancy estimation. The factors include the 5 cm cell size for row/alley detection, growth pattern of the plants, and wind condition during data acquisition. The *x* discrepancy estimated by different metrics are in agreement within a ±4 cm range. On the other hand, the variation in *y* discrepancy estimation ranges from 1 to 13 cm. This is because the *y* discrepancy was estimated based on the alley location, which is less reliable since it depends on when the planter starts/stops sowing. Another reason is that the number of alleys is fewer than the number of rows: for datasets B-1, B-2, and B-3, there were 90 rows and 13 alleys; for datasets C-1 and C-2, there were 260 rows and 21 alleys. Consequently, the redundancy for *y* discrepancy estimation is smaller. As an external check, the row spacing for each dataset was estimated using the row locations. The results are then compared to the row separation of the planter, which is 0.762 m in this study. Table 6 shows the statistics of the differences between the estimated row spacing and the row separation of the planter. The result shows that the estimated row spacing is consistent with the separation of the planter, with RMSE values in the range from 3 cm to 8 cm. In conclusion, the proposed horizontal quality control procedure provides a rough estimation of the

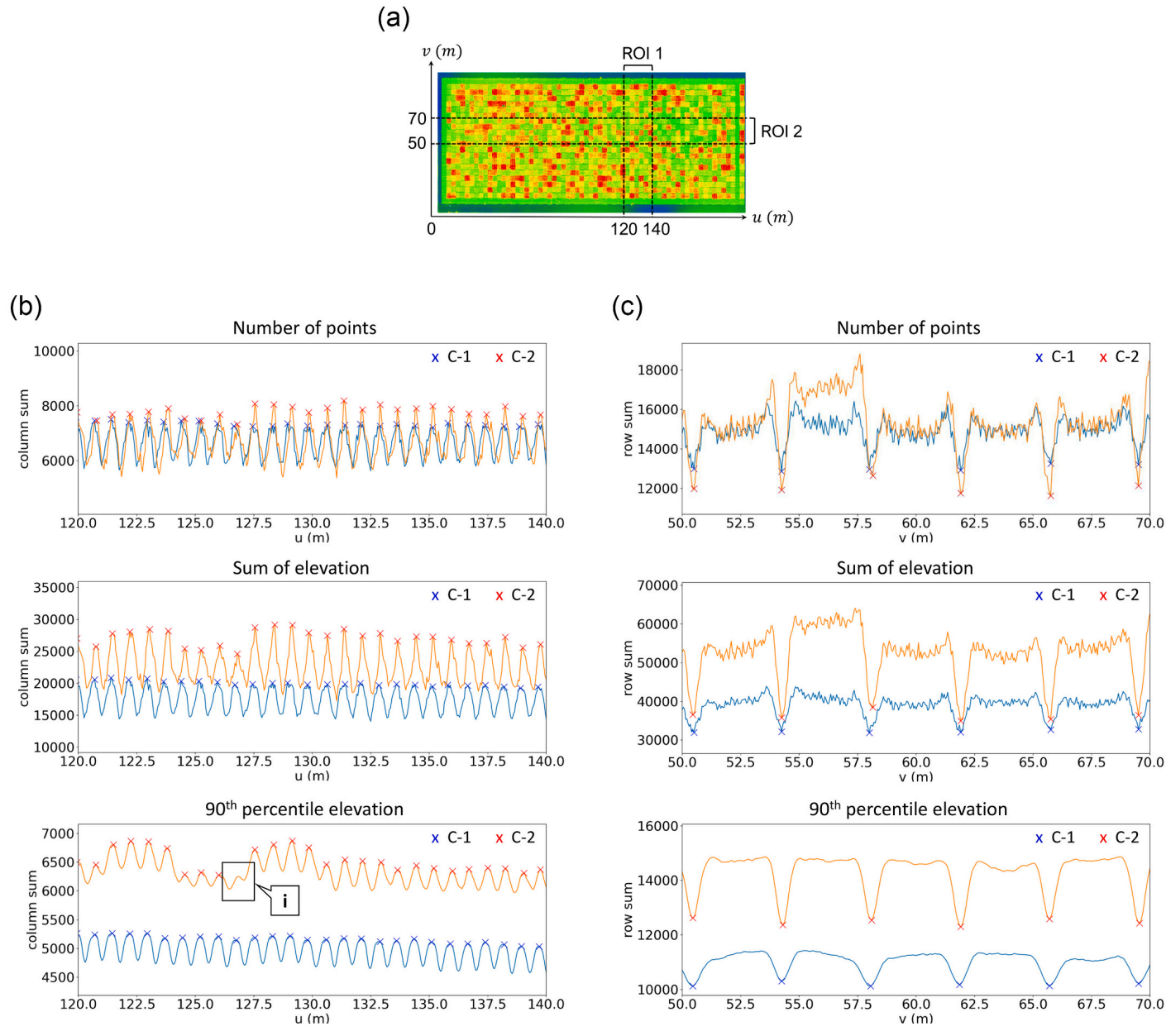

Fig. 16. Peak/valley detection with the three metrics for the C-1 and C-2 datasets: (a) point cloud and the selected ROIs (showing the C-2 dataset), (b) peak detection in ROI 1 where some misdetection can be observed at location *i*, and (c) valley detection in ROI 2.

Table 3
Performance of the three metrics for row/alley detection (estimated using B-1, B-2, B-3, C-1, and C-2 datasets).

Metric	Precision	Recall	F1-score
Number of points	99.1%	99.2%	99.2%
Sum of elevations	99.2%	99.8%	99.5%
90th percentile elevation	99.8%	98.8%	99.3%

planimetric accuracy. The estimated net discrepancies between multi-temporal point clouds are in a ±8 cm range. This discrepancy reflects a combined effect of i) accuracy of LiDAR point clouds, ii) cell size for row/alley detection, iii) crop type, iv) the growth pattern of the plants, and v) wind condition during data acquisition.

5.6. *Row/alley detection with non-orthogonal rows and alleys*

The robustness of the proposed row/alley detection strategy was further tested using the D-1 dataset. This maize field involved a distinct planting pattern where the rows and alleys are not orthogonal (i.e., the angle between the row and alley directions is 94°). Another challenge was that the field consists of two parts, and each part had different row segment lengths as mentioned in Section 3.2. Consequently, the alley locations for the two parts of the field were different. To tackle the issues mentioned above, we utilized the estimated row and alley directions, along with the prior information based on the field map that the first part of the field consists of 62 rows and the second part consists of 73 rows. First, the point cloud was rotated so that the rows are in the North-South direction. The row locations were identified by calculating the column sum using the three metrics and detecting the peaks. The location of the 62nd row was recorded as an indication of the boundary between the first and second parts of the field for alley detection. Then, the point cloud was rotated again to align the alley directions with the East-West direction. The row sum was calculated separately for each part of the field. Fig. 18 depicts the row and alley detection results using the three metrics for D-1 dataset. The result shows that by rotating the

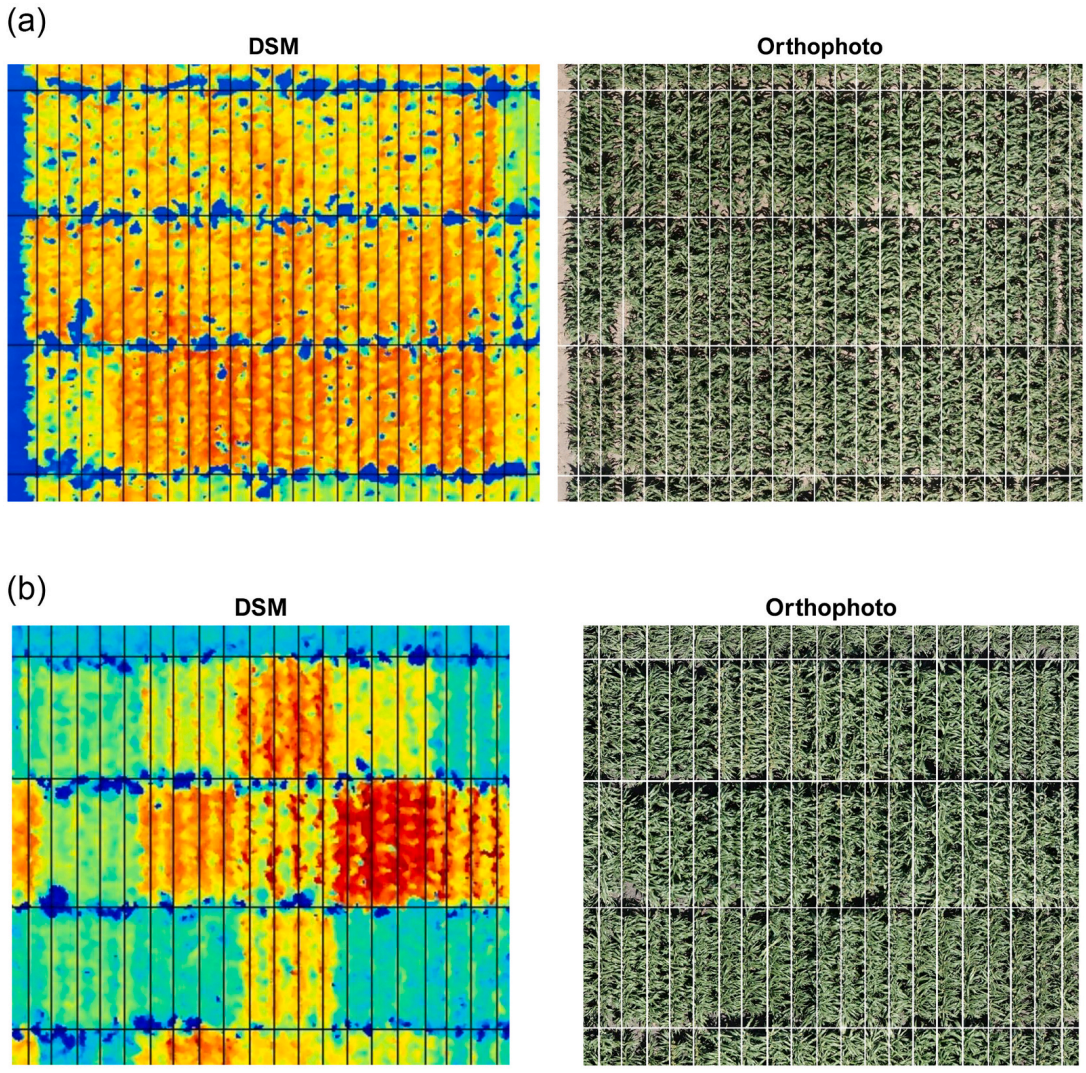

Fig. 17. Samples of the detected rows and alleys superimposed on top of the DSM and orthophoto showing a portion of the field for the (a) B-2, and (b) C-2 datasets.

Table 4
Estimated net discrepancy (d_x, d_y, and d_z) and the square root of a-posteriori variance factor ($\widehat{\sigma_0}$) using B-1, B-2, B-3, C-1, and C-2 datasets.

Reference dataset	Source dataset	$\widehat{\sigma_0}$ (m)	d_x (m)	d_y (m)	d_z (m)
B-1	B-2	0.02	0.04	0.11	0.01
B-1	B-3	0.03	0.01	0.04	−0.02
C-1	C-2	0.02	−0.12	−0.01	0.03

point cloud, the proposed algorithm was able to detect rows/alleys even though they were not orthogonal. The valleys based on the 90th percentile elevation seem to be less prominent in comparison to the valleys in the B-1, B-2, B-3, C-1, and C-2 datasets as the alleys were narrower. Fig. 19 shows the detected row and alley locations projected on top of the DSM and orthophoto. The detected rows and alleys are aligned well with the actual row centers and alleys shown in the DSM and orthophoto. The result demonstrates that we are able to detect the rows and alleys with LiDAR point clouds, while it is actually difficult to manually identify them due to the high planting density. Overall, the experimental results for the D-1 dataset show that the proposed strategy can deal with manually-planted fields, non-orthogonal rows and alleys, and high planting density as long as the crops are planted along straight lines.

6. Discussion

6.1. Performance of the quality control approaches

The performance of the proposed quality control framework was evaluated using multi-temporal datasets. The main findings of the

Table 5
Estimated net planimetric discrepancy (d_x and d_y) and the square root of a-posteriori variance factor ($\widehat{\sigma_0}$) based on the metrics: number of points, sum of elevation, and 90th percentile elevation using B-1, B-2, B-3, C-1, and C-2 datasets.

Reference dataset	Source dataset	Number of points			Sum of elevation			90th percentile elevation		
		$\widehat{\sigma_0}$ (m)	d_x (m)	d_y (m)	$\widehat{\sigma_0}$ (m)	d_x (m)	d_y (m)	$\widehat{\sigma_0}$ (m)	d_x (m)	d_y (m)
B-1	B-2	0.08	0.05	−0.06	0.07	0.04	0.02	0.07	0.01	0.07
B-1	B-3	0.08	0.08	0.02	0.09	0.08	0.07	0.07	0.07	0.07
C-1	C-2	0.07	−0.06	0.03	0.07	−0.07	0.03	0.05	−0.07	0.02

Table 6
Mean, standard deviation, and RMSE of the differences between the estimated row spacing and the row separation of the planter using B-1, B-2, B-3, C-1, and C-2 datasets.

Dataset	Mean (m)	Std. Dev. (m)	RMSE (m)
B-1	0.00	0.06	0.06
B-2	0.00	0.03	0.03
B-3	0.00	0.04	0.04
C-1	−0.01	0.08	0.08
C-2	0.00	0.05	0.05

analyses are as follows:

- The absolute accuracy of the point cloud from the UAV-based system is assessed against an independent RTK-GNSS survey. The result shows that the absolute accuracy of the point clouds is in the ±4 cm range in x, y, and z directions.
- The relative vertical discrepancies between multi-temporal point clouds were estimated using terrain patches. The result suggests that the net discrepancy is in the range of ±3 cm.
- The relative planimetric discrepancies between multi-temporal point clouds were estimated using the row and alley locations. The result suggests that the net discrepancy is in the range of ±8 cm.

The estimated relative vertical discrepancy is consistent with the absolute accuracy of the point clouds. This result indicates that the terrain is stable throughout the growing season despite the technical or environmental factors pertaining to the field trial or data collection. Therefore, the proposed approach can reliably determine the relative vertical accuracy of the point clouds as long as there is sufficient penetration of LiDAR beams to the ground. That is, the number of points

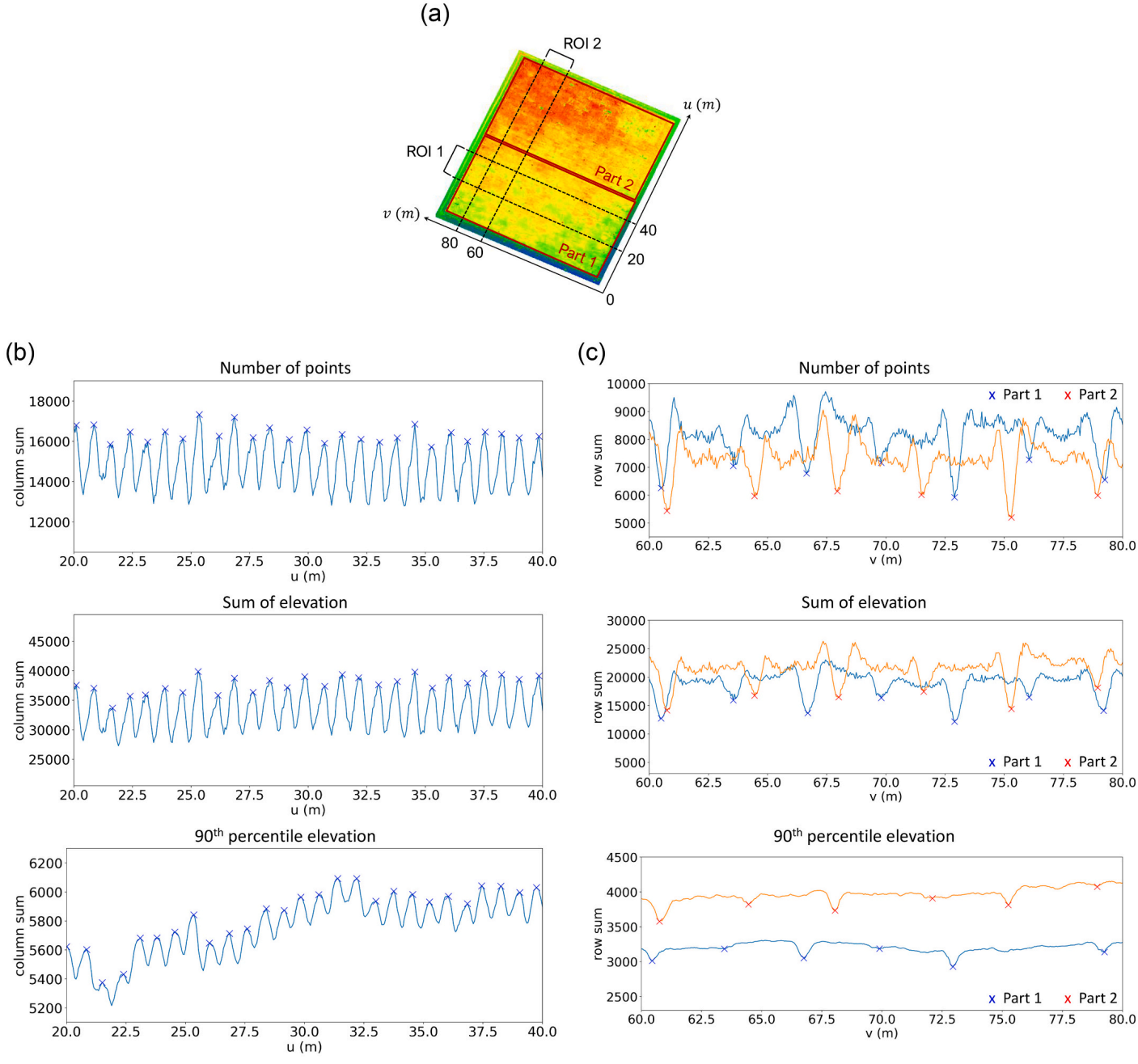

Fig. 18. Peak/valley detection with the three metrics for the D-1 dataset: (a) point cloud and the selected ROIs, (b) peak detection in ROI 1, and (c) valley detection in ROI 2.

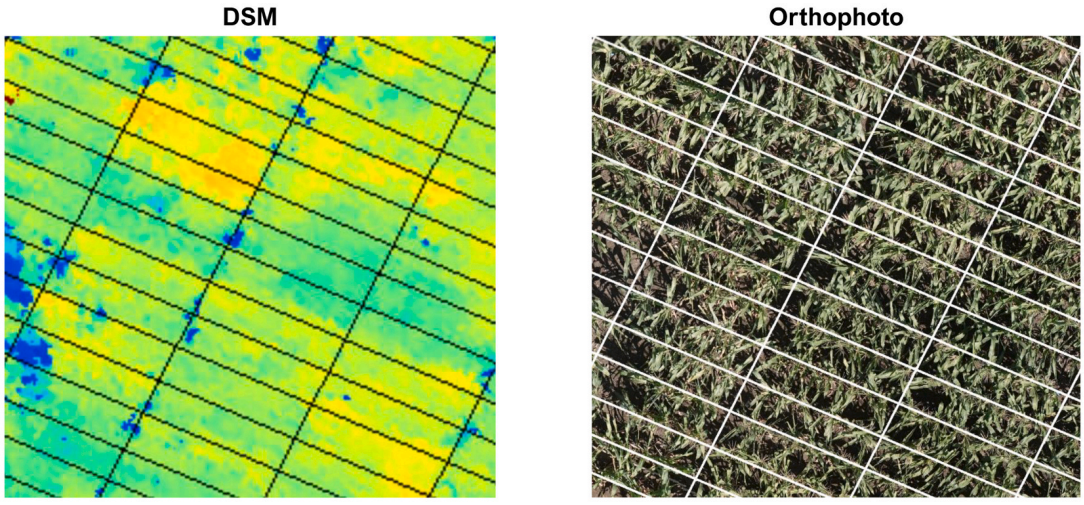

Fig. 19. Sample of the detected rows and alleys superimposed on top of the DSM and orthophoto showing a portion of the field using the D-1 dataset.

in each patch is large enough for determining the plane parameters.

On the other hand, the estimated relative planimetric discrepancy is slightly worse than the absolute accuracy of the point clouds. This result indicates that there are factors other than the accuracy of the point clouds that affect the discrepancy estimation. Several factors have been identified, including the cell size for row/alley detection, crop type, growth pattern of the plants, and wind condition during data acquisition. The selection of the cell size determines the precision of the detected row/alley locations. However, it has minor impact on the discrepancy estimation because of the large redundancy resulting from using all the detected rows and alleys in the least squares adjustment. The crop type and growth pattern have an influence on both the row/alley detection and discrepancy estimation. The experimental result shows that the dense canopy and large height variation among plots in the sorghum field might lead to some misdetection of rows/alleys. Moreover, some hybrids tend to lean as they grow taller. This will increase the uncertainty of the detected row/alley locations. The wind condition during data collection affects the noise level of the point clouds. In addition, if the wind direction is consistent throughout the data collection, it could result in having all plants tilted toward the same direction, and therefore introduce bias to the row/alley locations. Consequently, the planimetric discrepancy estimation reflects a combined effect of the point cloud accuracy as well as the above-mentioned factors. From a different point of view, deviations of the estimated relative accuracy from the expected values can be used to infer the growth pattern of the crop. One should note that these factors will not affect the vertical quality assessment since they are only related to row/alley locations.

6.2. Performance of the row/alley detection strategy and potential applications

A strategy for automated detection of rows and alleys is proposed and tested using six datasets with agricultural fields of different sizes, crops, planting patterns, planting densities, and growth stages. The key findings of the analyses are as follows:

- The proposed row/alley detection approach with either one of the three metrics – number of points, sum of elevations, and 90th percentile elevation – achieved high performance.
- The approach can deal with different crops, planting patterns (e.g., non-orthogonal rows and alleys), planting densities, and growth stages.

Once the row/alley locations are detected, individual row segments can be automatically extracted from LiDAR data. As discussed in Section 2, the first step of field phenotyping using remotely sensed data is to segment the data into individual plots, plants, or plant organs. LiDAR-based plot/plant segmentation relied heavily on manual work, pre-surveyed plot boundaries, or prior information regarding planting pattern (Su et al., 2019; Malambo et al., 2019; Qiu et al., 2019). Automated plot/plant extraction using image data usually works on early season, or low planting density areas where the distinction between ground and plants is clear (Ribera et al., 2017; Ahmed et al., 2019). Our approach can detect the row/alley locations in mid- or late-season with relative high planting density. Moreover, since the LiDAR datasets are georeferenced to a common mapping frame, the detected row segment boundaries can be superimposed onto field maps, orthophotos, or even back-projected to the raw images if the camera position and orientation information is known. Once the segments are extracted, plant traits such as height, volume, canopy cover, and LAI can be calculated for individual plots/plant varieties.

7. Conclusions and recommendations for future research

This paper presents a quality control procedure for multi-temporal LiDAR point clouds in mechanized agricultural fields. The key motivation of this study is to automatically extract features from agricultural fields, and achieve a targetless quality control procedure. Terrain patches extracted from the bare earth point clouds are utilized to evaluate the relative vertical accuracy, while row and alley locations in the field are used to assess the relative planimetric accuracy. The absolute accuracy of the point clouds generated from the UAV-based system was first assessed against RTK-GNSS survey. The results suggest that the absolute accuracy of the multi-temporal LiDAR point clouds from the UAV-based mobile mapping system is in the ±4 cm range. The performance of the proposed vertical and planimetric quality control framework was then validated using multi-temporal datasets collected in a maize field (datasets B-1, B-2, and B-3), and a sorghum field (datasets C-1 and C-2). The result shows that the net vertical and planimetric discrepancies between multi-temporal point clouds are ±3 cm and ±8 cm, respectively. While the former reflects the actual accuracy of the point clouds, the latter is a combined effect of the LiDAR point cloud accuracy, rasterization artifacts, crop type, growth pattern, and wind condition. In terms of row and alley detection, the result shows that either one of the metrics (number of points, sum of elevation, and 90th percentile elevation) can achieve high performance. The metric, sum of elevation, is the most robust metric since it considers the point density and height, and both characteristics are related to plant location. The row and alley detection result in Patancheru further demonstrates that the proposed

strategy is able to deal with non-orthogonal rows and alleys, dense canopy cover, and high planting density.

There are several potential directions for future research. Currently, the proposed approach is tested on maize and sorghum, which are major crops for food and biofuel production. We will apply the approach to other crops, for example, the legume family, to evaluate the versatility of this approach. The proposed row/alley detection extracts the plant locations as rows and alleys. We will explore the possibility of using LiDAR data to detect individual plant locations in the future. Our current work ensures the LiDAR data quality and extracts row segments automatically. The next step will be extracting plant traits at plot level.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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